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Dynamically Regularized Lagrange Multiplier Method for the Cahn-Hilliard-Navier-Stokes System

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ABSTRACT

This paper is concerned with efficient and accurate numerical schemes for the Cahn-Hilliard-Navier-Stokes phase field model of binary immiscible fluids. By introducing two Lagrange multipliers for each of the Cahn-Hilliard and Navier-Stokes parts, we reformulate the original model problem into an equivalent system that incorporates the energy evolution process. Such a nonlinear, coupled system is then discretized in time using first- and second-order backward differentiation formulas, in which all nonlinear terms are treated explicitly and no extra stabilization term is imposed. The proposed dynamically regularized Lagrange multiplier (DRLM) schemes are mass-conserving and unconditionally energy-stable with respect to the original variables. In addition, the schemes are fully decoupled: Each time step involves solving two biharmonic-type equations and two generalized linear Stokes systems, together with two nonlinear algebraic equations for the Lagrange multipliers. A key feature of the DRLM schemes is the introduction of the regularization parameters which ensure the unique determination of the Lagrange multipliers and mitigate the time step size constraint without affecting the accuracy of the numerical solution, especially when the interfacial width is small. Various numerical experiments are presented to illustrate the accuracy and robustness of the proposed DRLM schemes in terms of convergence, mass conservation, and energy stability.

1 | Introduction

The interface dynamics of binary immiscible fluids is an important topic in fluid mechanics with various applications in science and engineering. Mathematically, the interface between the two phases can be modeled as a moving boundary (i.e., sharp interface models [1–3]) or as a thin transition layer (i.e., diffuse interface models). The latter is also known as phase field models [4–8], in which a phase variable is introduced and takes distinct constant values in each bulk phase while varying

smoothly in the interfacial region. Based on an energetic variational approach, a single set of governing equations is derived on the whole domain (for both phases) and solved on a fixed mesh. Unlike the sharp interface models [9, 10], no explicit tracking of the interface is needed, and thus phase field models admit an energy law and provide an efficient approach for handling topological changes of the interface.

The Cahn-Hilliard-Navier-Stokes (CH-NS) system [11–15] is a well-known phase field model for the mixture of two

immiscible and incompressible Newtonian fluids with matched density. When the density difference is small, the model can also be applied with a Boussinesq approximation [16]. Numerical approximations for the nonlinear, coupled CH-NS system pose several challenges. First, to handle the stiffness of the phase equation associated with the interfacial width, numerical schemes are highly desired to be unconditionally stable. Second, the coupling of nonlinear equations introduces significant computational cost, particularly with implicit methods. Third, numerical solutions must preserve key physical principles, such as mass conservation and energy dissipation. During the past two decades, many studies have been carried out to develop and analyze unconditionally energy-stable schemes for the CH-NS equations. Popular methods include the convex splitting methods [17–21], stabilized semi-implicit methods [22–25], and scalar auxiliary variable (SAV) schemes [26–30]. Convex splitting schemes guarantee the unique solvability of the numerical solution, which is the minimizer of a convex functional [16]; however, a nonlinear equation needs to be solved at each time step. Stabilized schemes, on the other hand, introduce artificial numerical dissipation to the system, which potentially changes the system dynamics. An issue with SAV-based schemes is that the stability results often apply to a modified energy, which is not directly related to or close to the original energy when the time step size is not small enough. We refer to [31, 32] for other energy stable schemes for the CH-NS system, and to [33] for a bound-preserving limiter-based algorithm that employs discontinuous Galerkin discretization in space along with a combination of convex splitting and rotational pressure-correction techniques in time.

In this paper, we develop efficient and accurate numerical schemes for the CH-NS system to overcome the aforementioned challenges and drawbacks. Motivated by recent studies on the Lagrange multiplier approach for the gradient flows [34–36] and incompressible NS equations [37, 38], we first reformulate the CH-NS system into an equivalent form by introducing two Lagrange multipliers, one for the CH part of the system and the other for the NS part. A dynamic equation related to each of the two Lagrange multipliers is added to the system, involving a regularization parameter and the evolution of a certain component of the total energy associated with the CH equation or the NS equations. The coupled, nonlinear system is then discretized in time by first- and second-order backward differentiation formulas (BDF) with explicit treatment of all nonlinear terms, resulting in the so-called *dynamically regularized Lagrange multiplier* (DRLM) schemes.

The proposed DRLM schemes possess the following highly desirable properties: (i) there is no stabilization term in the formulation due to the introduction of the regularization parameters which ensure the unique determination of the Lagrange multipliers while mitigating strict time step size constraints typically imposed by explicit treatment of nonlinear terms (especially when the interfacial width is small); (ii) the schemes conserve mass and satisfy the unconditional energy stability with respect to the original variables; (iii) DRLM schemes are linear and fully decoupled, each time step requires the solution of two fourth-order linear equations involving the bilaplacian operator (for the phase variable), two generalized Stokes systems (for the fluid velocity and pressure fields), and two nonlinear algebraic

equations (for the Lagrange multipliers); and (iv) numerical results for various test cases, including shape relaxation, spinodal decomposition, and droplet dynamics, confirm the accuracy and stability of the DRLM schemes and their ability to capture the dynamics of two-phase systems.

The rest of the paper is organized as follows: In Section 2, we introduce the coupled CH-NS system and its equivalent formulation with the introduction of two Lagrange multipliers. Based on the BDF time discretization, first- and second-order DRLM schemes are constructed in Section 3, along with details about the numerical implementation of these schemes. In Section 4, we prove mass conservation and unconditional energy stability of the DRLM schemes. Different numerical experiments are carried out in Section 5 to verify the theoretical results as well as illustrate the accuracy and robustness of the DRLM schemes. Finally, some concluding remarks are given in Section 6.

2 | The Cahn-Hilliard-Navier-Stokes Model and Equivalent Formulation

Let $\Omega \subset \mathbb{R}^d$ ($d = 2, 3$) be an open and bounded domain, and $T > 0$ be the terminal time, we consider the following coupled Cahn-Hilliard-Navier-Stokes system:

$$\frac{\partial \phi}{\partial t} + \mathbf{u} \cdot \nabla \phi = M \Delta \mu, \quad \text{in } \Omega \times (0, T] \quad (1a)$$

$$\mu = \lambda(-\Delta \phi + F'(\phi)), \quad \text{in } \Omega \times (0, T] \quad (1b)$$

$$\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} - \nu \Delta \mathbf{u} + \nabla p = \mu \nabla \phi, \quad \text{in } \Omega \times (0, T] \quad (1c)$$

$$\nabla \cdot \mathbf{u} = 0, \quad \text{in } \Omega \times (0, T] \quad (1d)$$

subject to the initial conditions

$$\phi|_{t=0} = \phi_0, \quad \mathbf{u}|_{t=0} = \mathbf{u}_0, \quad \text{in } \Omega \quad (2)$$

and suitable boundary conditions. Throughout the paper, we consider either the periodic boundary conditions for all variables or the following no-flux/no-slip boundary conditions:

$$\frac{\partial \phi}{\partial \mathbf{n}} = \frac{\partial \mu}{\partial \mathbf{n}} = 0, \quad \mathbf{u} = 0, \quad \text{on } \partial\Omega \times (0, T] \quad (3)$$

where $\mathbf{n}$ denotes the outward unit normal vector to $\partial\Omega$. In Equation (1), the unknown quantities ϕ , μ , $\mathbf{u}$, and p represent the phase variable, the chemical potential, the velocity vector field, and the pressure, respectively. The given positive parameters are M (the mobility constant), λ (the mixing coefficient), and ν (the viscosity). The nonlinear free energy density is given by

$$F(\phi) = \frac{1}{4\epsilon^2}(\phi^2 - 1)^2$$

with ϵ denoting the thickness of the diffuse interface. The system (1) describes the motion of two incompressible, macroscopically immiscible Newtonian fluids with matched density (which is taken to be 1) and viscosity ν . We refer to [11, 12, 15] for the derivation of the model (and its extensions to other cases, for example, with variable density and variable viscosity) and to [39–41] for the existence and uniqueness of the solution to the model.

It is well known that the system (1) admits an energy dissipation law and is mass-conservative. In particular, the total energy associated with (1) is defined as

$$\begin{aligned} E(\phi, \mathbf{u}) &= E_{CH}(\phi) + E_{NS}(\mathbf{u}) \\ &= \lambda \int_{\Omega} \left(\frac{1}{2} |\nabla \phi|^2 + F(\phi) \right) d\mathbf{x} + \frac{1}{2} \int_{\Omega} |\mathbf{u}|^2 d\mathbf{x} \end{aligned} \quad (4)$$

By taking the L^2 inner product of (1a), (1b), and (1c) with μ , $\frac{\partial \phi}{\partial t}$, and $\mathbf{u}$, respectively, and using the incompressibility condition (1d), we obtain the energy dissipation law:

$$\frac{d}{dt} E(\phi, \mathbf{u}) = -M \|\nabla \mu\|^2 - \nu \|\nabla \mathbf{u}\|^2 \leq 0 \quad (5)$$

where $\|\cdot\|$ denotes the standard L^2 norm on Ω . The mass conservation property holds by taking the L^2 inner product of (1a) with 1 which yields

$$\frac{d}{dt} \int_{\Omega} \phi d\mathbf{x} = M \int_{\partial\Omega} \nabla \mu \cdot \mathbf{n} ds - \int_{\partial\Omega} \phi \mathbf{u} \cdot \mathbf{n} ds = 0 \quad (6)$$

where $\mathbf{u} \cdot \nabla \phi = \nabla \cdot (\phi \mathbf{u})$ due to (1d).

The goal of this work is to construct efficient, linear, decoupled numerical methods that preserve the two intrinsic properties (5) and (6) at the discrete level and accurately capture the dynamics of the model, especially when ε is small. To that end, by introducing two time-dependent Lagrange multipliers $\eta(t)$ and $q(t)$ with $\eta(0) = q(0) = 1$, we reformulate the CH-NS model (1) into the following system, subject to the same initial and boundary conditions:

$$\frac{\partial \phi}{\partial t} + \eta \mathbf{u} \cdot \nabla \phi = M \Delta \mu, \quad \text{in } \Omega \times (0, T] \quad (7a)$$

$$\mu = \lambda (-\Delta \phi + \eta F'(\phi)), \quad \text{in } \Omega \times (0, T] \quad (7b)$$

$$\frac{\partial \mathbf{u}}{\partial t} + q \mathbf{u} \cdot \nabla \mathbf{u} - \nu \Delta \mathbf{u} + \nabla p = \eta \mu \nabla \phi, \quad \text{in } \Omega \times (0, T] \quad (7c)$$

$$\nabla \cdot \mathbf{u} = 0, \quad \text{in } \Omega \times (0, T] \quad (7d)$$

$$\frac{d}{dt} \int_{\Omega} F(\phi) d\mathbf{x} + \theta_{\eta} \frac{d\eta^2}{dt} = \eta \int_{\Omega} F'(\phi) \frac{\partial \phi}{\partial t} d\mathbf{x}, \quad \text{in } (0, T] \quad (7e)$$

$$\begin{aligned} &\frac{d}{dt} \int_{\Omega} \frac{1}{2} |\mathbf{u}|^2 d\mathbf{x} + \theta_q \frac{dq^2}{dt} \\ &= \int_{\Omega} [\mathbf{u}_t + q(\mathbf{u} \cdot \nabla \mathbf{u})] \cdot \mathbf{u} d\mathbf{x} \\ &\quad + \eta \int_{\Omega} [(\mathbf{u} \cdot \nabla \phi) \mu - (\mu \nabla \phi) \cdot \mathbf{u}] d\mathbf{x}, \quad \text{in } (0, T] \end{aligned} \quad (7f)$$

where θ_{η} and θ_q are two positive regularization parameters associated with the CH and NS parts, respectively. We remark that the role of η and q is to enforce dissipation of the total energy while the regularization parameters θ_{η} and θ_q are crucial for relaxing time step size constraints caused by explicit treatment of nonlinear functions in the proposed DRLM schemes derived later in the next section. Moreover, θ_q guarantees the uniqueness of the Lagrange multiplier q , so that the following lemma holds.

Lemma 1. *The system (7) is equivalent to the original CH-NS model (1).*

Proof. If $\phi, \mu, \mathbf{u}$, and p solve the original model (1), then they, together with $\eta \equiv 1$ and $q \equiv 1$, satisfy the reformulated system (7).

Conversely, if (7) admits a solution $\phi, \mu, \mathbf{u}, p, \eta$, and q , we observe that

$$\int_{\Omega} (\mathbf{u} \cdot \nabla \mathbf{u}) \cdot \mathbf{u} d\mathbf{x} = 0 \quad (8)$$

$$\int_{\Omega} [(\mathbf{u} \cdot \nabla \phi) \mu - (\mu \nabla \phi) \cdot \mathbf{u}] d\mathbf{x} = 0 \quad (9)$$

From Equations (7f), (8), and (9), along with $q(0) = 1$, we find that $q(t) \equiv 1$ for all $t > 0$. On the other hand, (7e) can be simplified as

$$\eta \frac{d\eta}{dt} = \alpha(\eta - 1)$$

where $\alpha = \frac{1}{2\theta_{\eta}} \frac{d}{dt} \int_{\Omega} F(\phi) d\mathbf{x}$ is a time-dependent function. It follows that

$$\begin{aligned} &\frac{d}{dt} \left[(\eta(t) - 1) \exp \left(\eta(t) - \int_0^t \alpha(s) ds \right) \right] \\ &= \exp \left(\eta(t) - \int_0^t \alpha(s) ds \right) \left[\frac{d\eta(t)}{dt} + (\eta(t) - 1) \left(\frac{d\eta(t)}{dt} - \alpha(t) \right) \right] \\ &= \exp \left(\eta(t) - \int_0^t \alpha(s) ds \right) \left[\eta(t) \frac{d\eta(t)}{dt} - \alpha(t)(\eta(t) - 1) \right] = 0 \end{aligned} \quad (10)$$

Since $\eta(0) = 1$, we obtain from Equation (10) that $\eta(t) \equiv 1$ for all $t > 0$. Consequently, $\phi, \mu, \mathbf{u}$, and p satisfy the system (1), which completes the proof of the lemma. $\square$

3 | Dynamically Regularized Lagrange Multiplier Schemes

We consider a uniform partition of the time interval $[0, T]$: $0 = t_0 < t_1 < \dots < t_N = T$ with the time step size $\tau = \frac{T}{N}$ for the time discretization of the system (7) with the two Lagrange multipliers. In the following, we derive the first- and second-order dynamically regularized Lagrange multiplier (DRLM) schemes, along with their efficient numerical implementation.

3.1 | The First-Order DRLM Scheme

For $\psi \in \{\phi, \mu, \mathbf{u}, p, \eta, q\}$ and $0 \leq n \leq N$, let ψ^n be an approximation of $\psi(t_n)$. By applying the first-order BDF (i.e., backward Euler) scheme to the system (7) with explicit treatment of the coupling term $\mathbf{u} \cdot \nabla \phi$ and nonlinear terms $F'(\phi)$ and $\mathbf{u} \cdot \nabla \mathbf{u}$, we obtain the following first-order (in time) DRLM scheme: For $0 \leq n \leq N - 1$,

$$\frac{\phi^{n+1} - \phi^n}{\tau} + \eta^{n+1} \mathbf{u}^n \cdot \nabla \phi^n = M \Delta \mu^{n+1} \quad (11a)$$

$$\mu^{n+1} = \lambda (-\Delta \phi^{n+1} + \eta^{n+1} F'(\phi^n)) \quad (11b)$$

$$\frac{\mathbf{u}^{n+1} - \mathbf{u}^n}{\tau} + q^{n+1} \mathbf{u}^n \cdot \nabla \mathbf{u}^n - \nu \Delta \mathbf{u}^{n+1} + \nabla p^{n+1} = \eta^{n+1} \mu^{n+1} \nabla \phi^{n+1} \quad (11c)$$

$$\nabla \cdot \mathbf{u}^{n+1} = 0 \quad (11d)$$

$$\begin{aligned} &\int_{\Omega} [F(\phi^{n+1}) - F(\phi^n)] d\mathbf{x} + \theta_{\eta} [(\eta^{n+1})^2 - (\eta^n)^2] \\ &= \eta^{n+1} \int_{\Omega} F'(\phi^n) (\phi^{n+1} - \phi^n) d\mathbf{x} \end{aligned} \quad (11e)$$

$$\begin{aligned} & \int_{\Omega} \frac{|\mathbf{u}^{n+1}|^2 - |\mathbf{u}^n|^2}{2} d\mathbf{x} + \theta_q [(q^{n+1})^2 - (q^n)^2] \\ &= \int_{\Omega} (\mathbf{u}^{n+1} - \mathbf{u}^n + \tau q^{n+1} \mathbf{u}^n \cdot \nabla \mathbf{u}^n) \cdot \mathbf{u}^{n+1} d\mathbf{x} \\ &+ \tau \eta^{n+1} \int_{\Omega} [(\mathbf{u}^n \cdot \nabla \phi^n) \mu^{n+1} - (\mu^{n+1} \nabla \phi^{n+1}) \cdot \mathbf{u}^{n+1}] d\mathbf{x} \end{aligned} \quad (11f)$$

We note that the right-hand side of (7c) is treated implicitly in Equation (11c) because the CH variables at t_{n+1} , namely ϕ^{n+1} , μ^{n+1} , and η^{n+1} , can be computed from Equations (11a), (11b), and (11e) before solving the NS Equations (11c), (11d), and (11f) for $\mathbf{u}^{n+1}$, p^{n+1} , and q^{n+1} . In particular, let us decompose each of the unknown quantities ϕ^{n+1} , μ^{n+1} , $\mathbf{u}^{n+1}$, and p^{n+1} into two components as follows:

$$\phi^{n+1} = \phi_1^{n+1} + \eta^{n+1} \phi_2^{n+1} \quad (12a)$$

$$\mu^{n+1} = \mu_1^{n+1} + \eta^{n+1} \mu_2^{n+1} \quad (12b)$$

$$\mathbf{u}^{n+1} = \mathbf{u}_1^{n+1} + q^{n+1} \mathbf{u}_2^{n+1} \quad (12c)$$

$$p^{n+1} = p_1^{n+1} + q^{n+1} p_2^{n+1} \quad (12d)$$

Plugging (12a–12b) into (11a–11b), we obtain the following two linear equations in Ω for $(\phi_i^{n+1}, \mu_i^{n+1})$, $i = 1, 2$:

$$\frac{\phi_1^{n+1} - \phi^n}{\tau} = M \Delta \mu_1^{n+1}, \quad \mu_1^{n+1} = -\lambda \Delta \phi_1^{n+1} \quad (13)$$

$$\frac{\phi_2^{n+1}}{\tau} + \mathbf{u}^n \cdot \nabla \phi^n = M \Delta \mu_2^{n+1}, \quad \mu_2^{n+1} = \lambda (-\Delta \phi_2^{n+1} + F'(\phi^n)) \quad (14)$$

where ϕ_i^{n+1} and μ_i^{n+1} ($i = 1, 2$) satisfy the same boundary conditions as ϕ^{n+1} and μ^{n+1} , respectively, that is, either periodic or no-flux boundary conditions.

With $(\phi_i^{n+1}, \mu_i^{n+1})$, $i = 1, 2$, determined from the above Equations (13) and (14), we compute the Lagrange multiplier η^{n+1} from Equation (11e) by solving the following nonlinear algebraic equation:

$$\int_{\Omega} F(\phi_1^{n+1} + \eta^{n+1} \phi_2^{n+1}) d\mathbf{x} + A_{1,n+1}^{\eta} (\eta^{n+1})^2 + B_{1,n+1}^{\eta} \eta^{n+1} + C_{1,n+1}^{\eta} = 0 \quad (15)$$

where the quadratic coefficients are given by

$$\begin{aligned} A_{1,n+1}^{\eta} &= \theta_{\eta} - \int_{\Omega} F'(\phi^n) \phi_2^{n+1} d\mathbf{x}, \\ B_{1,n+1}^{\eta} &= - \int_{\Omega} F'(\phi^n) (\phi_1^{n+1} - \phi^n) d\mathbf{x}, \\ C_{1,n+1}^{\eta} &= -\theta_{\eta} (\eta^n)^2 - \int_{\Omega} F(\phi^n) d\mathbf{x} \end{aligned}$$

The complexity of the nonlinear Equation (15) depends on the potential function $F(\phi)$. In this paper, we focus on the commonly used double-well potential; hence, (15) is a fourth-order algebraic equation. Given that the exact value of η is 1, we solve (15) using Newton's iterative method with an initial guess of 1. Once η^{n+1} is known, we update ϕ^{n+1} and μ^{n+1} from Equations (12a) and (12b), respectively.

Regarding the NS equations, substituting (12c–12d) into (11c–11d) yields the following two linear systems of generalized Stokes equations in Ω for $(\mathbf{u}_i^{n+1}, p_i^{n+1})$, $i = 1, 2$:

$$\frac{\mathbf{u}_1^{n+1} - \mathbf{u}^n}{\tau} - \nu \Delta \mathbf{u}_1^{n+1} + \nabla p_1^{n+1} = \eta^{n+1} \mu^{n+1} \nabla \phi^{n+1}, \quad \nabla \cdot \mathbf{u}_1^{n+1} = 0 \quad (16)$$

$$\frac{\mathbf{u}_2^{n+1}}{\tau} + \mathbf{u}^n \cdot \nabla \mathbf{u}^n - \nu \Delta \mathbf{u}_2^{n+1} + \nabla p_2^{n+1} = 0, \quad \nabla \cdot \mathbf{u}_2^{n+1} = 0 \quad (17)$$

where $\mathbf{u}_1^{n+1}$ and $\mathbf{u}_2^{n+1}$ satisfy the same boundary conditions as $\mathbf{u}^{n+1}$, that is, either periodic or no-slip boundary conditions.

Once $\mathbf{u}_i^{n+1}$ and p_i^{n+1} ($i = 1, 2$) are determined from the above systems (16) and (17), the Lagrange multiplier q^{n+1} is computed from Equation (11f) by solving the quadratic equation:

$$A_{1,n+1}^q (q^{n+1})^2 + B_{1,n+1}^q q^{n+1} + C_{1,n+1}^q = 0 \quad (18)$$

where the quadratic coefficients are defined as:

$$\begin{aligned} A_{1,n+1}^q &= \theta_q + \frac{1}{2} \|\mathbf{u}_2^{n+1}\|^2 + \tau \nu \|\nabla \mathbf{u}_2^{n+1}\|^2, \\ B_{1,n+1}^q &= - \int_{\Omega} (\mathbf{u}_1^{n+1} - \mathbf{u}^n) \cdot \mathbf{u}_2^{n+1} d\mathbf{x} + \tau \int_{\Omega} [\eta^{n+1} (\mu^{n+1} \nabla \phi^{n+1}) \\ &\quad \cdot \mathbf{u}_2^{n+1} - (\mathbf{u}^n \cdot \nabla \mathbf{u}^n) \cdot \mathbf{u}_1^{n+1}] d\mathbf{x}, \\ C_{1,n+1}^q &= -\theta_q (q^n)^2 - \frac{1}{2} \|\mathbf{u}_1^{n+1} - \mathbf{u}^n\|^2 - \tau \eta^{n+1} \\ &\quad \times \int_{\Omega} [(\mathbf{u}^n \cdot \nabla \phi^n) \mu^{n+1} - (\mu^{n+1} \nabla \phi^{n+1}) \cdot \mathbf{u}_1^{n+1}] d\mathbf{x} \end{aligned}$$

Note that in the derivation of $A_{1,n+1}^q$, we have used the fact that

$$\tau \int_{\Omega} (\mathbf{u}^n \cdot \nabla \mathbf{u}^n) \cdot \mathbf{u}_2^{n+1} d\mathbf{x} = -\|\mathbf{u}_2^{n+1}\|^2 - \tau \nu \|\nabla \mathbf{u}_2^{n+1}\|^2$$

obtained from Equation (17) and integration by parts. Since $A_{1,n+1}^q > 0$ for all $\theta_q > 0$, (18) has two solutions, and q^{n+1} is chosen as the one closer to 1. Finally, we update $\mathbf{u}^{n+1}$ and p^{n+1} via (12c) and (12d), respectively.

3.2 | The Second-Order DRLM Scheme

Let us denote by $\tilde{\phi}^{n+1} = 2\phi^n - \phi^{n-1}$ and $\tilde{\mathbf{u}}^{n+1} = 2\mathbf{u}^n - \mathbf{u}^{n-1}$ the second-order extrapolations to approximate $\mathbf{u} \cdot \nabla \phi$, $F'(\phi)$, and $\mathbf{u} \cdot \nabla \mathbf{u}$ at t_{n+1} . Applying the BDF2 scheme to the system (7) yields the following second-order (in time) DRLM scheme: for $1 \leq n \leq N-1$,

$$\frac{3\phi^{n+1} - 4\phi^n + \phi^{n-1}}{2\tau} + \eta^{n+1} \tilde{\mathbf{u}}^{n+1} \cdot \nabla \tilde{\phi}^{n+1} = M \Delta \mu^{n+1} \quad (19a)$$

$$\mu^{n+1} = \lambda (-\Delta \phi^{n+1} + \eta^{n+1} F'(\tilde{\phi}^{n+1})) \quad (19b)$$

$$\frac{3\mathbf{u}^{n+1} - 4\mathbf{u}^n + \mathbf{u}^{n-1}}{2\tau} + q^{n+1} \tilde{\mathbf{u}}^{n+1} \cdot \nabla \tilde{\mathbf{u}}^{n+1} - \nu \Delta \mathbf{u}^{n+1} + \nabla p^{n+1} = \eta^{n+1} \mu^{n+1} \nabla \phi^{n+1} \quad (19c)$$

$$\nabla \cdot \mathbf{u}^{n+1} = 0 \quad (19d)$$

$$\begin{aligned} & \int_{\Omega} [3F(\phi^{n+1}) - 4F(\phi^n) + F(\phi^{n-1})] d\mathbf{x} + \theta_{\eta} [3(\eta^{n+1})^2 - 4(\eta^n)^2 + (\eta^{n-1})^2] \\ &= \eta^{n+1} \int_{\Omega} F'(\tilde{\phi}^{n+1}) (3\phi^{n+1} - 4\phi^n + \phi^{n-1}) d\mathbf{x} \quad (19e) \\ & \int_{\Omega} \frac{3|\mathbf{u}^{n+1}|^2 - 4|\mathbf{u}^n|^2 + |\mathbf{u}^{n-1}|^2}{2} d\mathbf{x} + \theta_q [3(q^{n+1})^2 - 4(q^n)^2 + (q^{n-1})^2] \end{aligned}$$

$$= \int_{\Omega} (3\mathbf{u}^{n+1} - 4\mathbf{u}^n + \mathbf{u}^{n-1} + 2\tau q^{n+1} \tilde{\mathbf{u}}^{n+1} \cdot \nabla \tilde{\mathbf{u}}^{n+1}) \cdot \mathbf{u}^{n+1} d\mathbf{x} \\ + 2\tau \eta^{n+1} \int_{\Omega} [(\tilde{\mathbf{u}}^{n+1} \cdot \nabla \tilde{\phi}^{n+1}) \mu^{n+1} - (\mu^{n+1} \nabla \phi^{n+1}) \cdot \mathbf{u}^{n+1}] d\mathbf{x} \quad (19f)$$

To initialize the second-order DRLM scheme (19), we use $\{\phi^0, \mathbf{u}^0, \eta^0 = 1, q^0 = 1\}$ from the initial conditions (2) and $\{\phi^1, \mathbf{u}^1, \eta^1, q^1\}$ computed by the first-order DRLM scheme (11). It should be noted that the pressure p and chemical potential μ at t_{n-1} and t_n are not required to compute the solution $\{\phi^{n+1}, \mu^{n+1}, \eta^{n+1}, \mathbf{u}^{n+1}, p^{n+1}, q^{n+1}\}$ at time t_{n+1} . We now describe how to efficiently implement the second-order DRLM scheme (19). Similarly to the first-order case, we again consider the decomposition (12) for the unknown quantities ϕ^{n+1} , μ^{n+1} , $\mathbf{u}^{n+1}$, and p^{n+1} . By substituting (12a–12b) into (19a–19b), we get the following two linear equations for solving $(\phi_i^{n+1}, \mu_i^{n+1})$, $i = 1, 2$:

$$\frac{3\phi_1^{n+1} - 4\phi^n + \phi^{n-1}}{2\tau} = M \Delta \mu_1^{n+1}, \quad \mu_1^{n+1} = -\lambda \Delta \phi_1^{n+1} \quad (20)$$

$$\frac{3\phi_2^{n+1}}{2\tau} + \tilde{\mathbf{u}}^{n+1} \cdot \nabla \tilde{\phi}^{n+1} = M \Delta \mu_2^{n+1}, \quad \mu_2^{n+1} = \lambda (-\Delta \phi_2^{n+1} + F'(\tilde{\phi}^{n+1})) \quad (21)$$

Next, we update the Lagrange multiplier η^{n+1} from Equation (19e), which can be written as:

$$3 \int_{\Omega} F(\phi_1^{n+1} + \eta^{n+1} \phi_2^{n+1}) d\mathbf{x} + A_{2,n+1}^{\eta} (\eta^{n+1})^2 + B_{2,n+1}^{\eta} \eta^{n+1} + C_{2,n+1}^{\eta} = 0 \quad (22)$$

where

$$A_{2,n+1}^{\eta} = 3\theta_{\eta} - 3 \int_{\Omega} F'(\tilde{\phi}^{n+1}) \phi_2^{n+1} d\mathbf{x},$$

$$B_{2,n+1}^{\eta} = - \int_{\Omega} F'(\tilde{\phi}^{n+1}) (3\phi_1^{n+1} - 4\phi^n + \phi^{n-1}) d\mathbf{x},$$

$$C_{2,n+1}^{\eta} = \theta_{\eta} [-4(\eta^n)^2 + (\eta^{n-1})^2] + \int_{\Omega} [-4F(\phi^n) + F(\phi^{n-1})] d\mathbf{x}$$

Note that (22) can be solved iteratively using Newton's method with an initial guess of 1. After determining η^{n+1} from Equation (22), we update ϕ^{n+1} and μ^{n+1} according to (12a) and (12b), respectively. The combination of (12c–12d) and (19c–19d) results in the following two linear Stokes systems for $(\mathbf{u}_i^{n+1}, p_i^{n+1})$, $i = 1, 2$:

$$\frac{3\mathbf{u}_1^{n+1} - 4\mathbf{u}^n + \mathbf{u}^{n-1}}{2\tau} - \nu \Delta \mathbf{u}_1^{n+1} + \nabla p_1^{n+1} = \eta^{n+1} \mu^{n+1} \nabla \phi^{n+1} \quad \nabla \cdot \mathbf{u}_1^{n+1} = 0 \quad (23)$$

$$\frac{3\mathbf{u}_2^{n+1}}{2\tau} + \tilde{\mathbf{u}}^{n+1} \cdot \nabla \tilde{\mathbf{u}}^{n+1} - \nu \Delta \mathbf{u}_2^{n+1} + \nabla p_2^{n+1} = 0 \quad \nabla \cdot \mathbf{u}_2^{n+1} = 0 \quad (24)$$

Given $(\mathbf{u}_i^{n+1}, p_i^{n+1})$ for $i = 1, 2$ from Equations (23, 24) and simplifying (19f), we find that q^{n+1} satisfies the following quadratic equation:

$$A_{2,n+1}^q (q^{n+1})^2 + B_{2,n+1}^q q^{n+1} + C_{2,n+1}^q = 0 \quad (25)$$

where

$$A_{2,n+1}^q = 3\theta_q + \frac{3}{2} \|\mathbf{u}_2^{n+1}\|^2 + 2\tau \nu \|\nabla \mathbf{u}_2^{n+1}\|^2,$$

$$B_{2,n+1}^q = - \int_{\Omega} (3\mathbf{u}_1^{n+1} - 4\mathbf{u}^n + \mathbf{u}^{n-1}) \cdot \mathbf{u}_2^{n+1} d\mathbf{x} \\ + 2\tau \int_{\Omega} [\eta^{n+1} (\mu^{n+1} \nabla \phi^{n+1}) \cdot \mathbf{u}_2^{n+1} - (\tilde{\mathbf{u}}^{n+1} \cdot \nabla \tilde{\mathbf{u}}^{n+1}) \cdot \mathbf{u}_1^{n+1}] d\mathbf{x},$$

$$C_{2,n+1}^q = \theta_q [-4(q^n)^2 + (q^{n-1})^2] - 2\|\mathbf{u}_1^{n+1} - \mathbf{u}^n\|^2 + \frac{1}{2} \|\mathbf{u}_1^{n+1} - \mathbf{u}^{n-1}\|^2 \\ - 2\tau \eta^{n+1} \int_{\Omega} [(\tilde{\mathbf{u}}^{n+1} \cdot \nabla \tilde{\phi}^{n+1}) \mu^{n+1} - (\mu^{n+1} \nabla \phi^{n+1}) \cdot \mathbf{u}_1^{n+1}] d\mathbf{x}$$

To derive $A_{2,n+1}^q$, we have used (24) and integration by parts. Since $A_{2,n+1}^q > 0$ for all $\theta_q > 0$, Equation (25) has two solutions, with q^{n+1} selected as the one closer to 1. Finally, we determine $\mathbf{u}^{n+1}$ and p^{n+1} from Equations (12c) and (12d), respectively.

4 | Properties of the Time-Discrete Solutions

In the following, the first- and second-order DRLM schemes (cf. (11) and (19), respectively) are shown to be mass-conserving and unconditionally energy stable. We remark that, differently from existing studies on SAV-based schemes, our energy stability results are established with respect to the original variables rather than the modified ones.

4.1 | Mass Conservation

We verify that, under the prescribed boundary conditions, the proposed time-discrete DRLM schemes conserve mass over time.

Theorem 1. *The proposed first-order (11) and second-order (19) DRLM schemes satisfy mass conservation:*

$$\int_{\Omega} \phi^{n+1} d\mathbf{x} = \int_{\Omega} \phi^n d\mathbf{x}, \quad n = 0, 1, \dots, N-1 \quad (26)$$

Proof. For the first-order scheme (11), taking the integral of (11a) over Ω gives us

$$\frac{1}{\tau} \int_{\Omega} (\phi^{n+1} - \phi^n) d\mathbf{x} + \eta^{n+1} \int_{\Omega} \mathbf{u}^n \cdot \nabla \phi^n d\mathbf{x} = M \int_{\Omega} \Delta \mu^{n+1} d\mathbf{x} \quad (27)$$

Since μ^{n+1} satisfies either periodic or homogeneous Neumann boundary conditions, integration by parts yields

$$\int_{\Omega} \Delta \mu^{n+1} d\mathbf{x} = \int_{\partial\Omega} \nabla \mu^{n+1} \cdot \mathbf{n} ds = 0 \quad (28)$$

In light of the incompressibility condition (11d) and the boundary conditions of $\mathbf{u}^n$ and ϕ^n , we observe that

$$\int_{\Omega} \mathbf{u}^n \cdot \nabla \phi^n d\mathbf{x} = \int_{\Omega} \nabla \cdot (\phi^n \mathbf{u}^n) d\mathbf{x} = \int_{\partial\Omega} (\phi^n \mathbf{u}^n) \cdot \mathbf{n} ds = 0 \quad (29)$$

The combination of (27–29) results in

$$\int_{\Omega} (\phi^{n+1} - \phi^n) d\mathbf{x} = 0$$

which indicates the mass conservation of the first-order scheme (11).

For the second-order counterpart (19), it is clear that (26) holds for $n = 0$. The proof is completed by induction, noting from Equation (19a) that

$$\begin{aligned} & \frac{1}{2\tau} \int_{\Omega} (3\phi^{n+1} - 4\phi^n + \phi^{n-1}) d\mathbf{x} \\ &= M \int_{\Omega} \Delta \mu^{n+1} d\mathbf{x} - \eta^{n+1} \int_{\Omega} \nabla \cdot (\tilde{\phi}^{n+1} \tilde{\mathbf{u}}^{n+1}) d\mathbf{x} = 0 \end{aligned}$$

where we have used the fact that $\tilde{\mathbf{u}}^{n+1} = 2\mathbf{u}^n - \mathbf{u}^{n-1}$ is divergence-free. $\square$

4.2 | Energy Stability

Our goal in this subsection is to show that the two proposed DRLM schemes satisfy energy stability with respect to the original energy functional (4).

Theorem 2. *The first-order DRLM scheme (11) is unconditionally energy stable in the sense that for $0 \leq n \leq N-1$, we have*

$$E(\phi^{n+1}, \mathbf{u}^{n+1}) + \mathcal{E}_1(\eta^{n+1}, q^{n+1}) \leq E(\phi^n, \mathbf{u}^n) + \mathcal{E}_1(\eta^n, q^n) \quad (30)$$

where

$$\mathcal{E}_1(\eta^n, q^n) := \lambda \theta_{\eta} [(\eta^n)^2 - 1] + \theta_q [(q^n)^2 - 1]$$

Proof. Taking the L^2 inner product of (11b) with $\phi^{n+1} - \phi^n$ and (11a) with $\tau \mu^{n+1}$ yields

$$\begin{aligned} & \lambda \int_{\Omega} (-\Delta \phi^{n+1} + \eta^{n+1} F'(\phi^n)) (\phi^{n+1} - \phi^n) d\mathbf{x} \\ &= \int_{\Omega} \mu^{n+1} (\phi^{n+1} - \phi^n) d\mathbf{x} \\ &= -\tau \eta^{n+1} \int_{\Omega} (\mathbf{u}^n \cdot \nabla \phi^n) \mu^{n+1} d\mathbf{x} - \tau M \|\nabla \mu^{n+1}\|^2 \\ &\leq -\tau \eta^{n+1} \int_{\Omega} (\mathbf{u}^n \cdot \nabla \phi^n) \mu^{n+1} d\mathbf{x} \end{aligned} \quad (31)$$

It follows from integration by parts that

$$\begin{aligned} & \int_{\Omega} (-\Delta \phi^{n+1}) (\phi^{n+1} - \phi^n) d\mathbf{x} \\ &= \int_{\Omega} \nabla \phi^{n+1} \cdot \nabla (\phi^{n+1} - \phi^n) d\mathbf{x} \\ &= \frac{1}{2} (\|\nabla \phi^{n+1}\|^2 - \|\nabla \phi^n\|^2) + \frac{1}{2} \|\nabla (\phi^{n+1} - \phi^n)\|^2 \\ &\geq \frac{1}{2} (\|\nabla \phi^{n+1}\|^2 - \|\nabla \phi^n\|^2) \end{aligned} \quad (32)$$

The combination of (31) and (32) results in

$$\begin{aligned} & \lambda \left[\frac{1}{2} (\|\nabla \phi^{n+1}\|^2 - \|\nabla \phi^n\|^2) + \eta^{n+1} \int_{\Omega} F'(\phi^n) (\phi^{n+1} - \phi^n) d\mathbf{x} \right] \\ &\leq -\tau \eta^{n+1} \int_{\Omega} (\mathbf{u}^n \cdot \nabla \phi^n) \mu^{n+1} d\mathbf{x} \end{aligned}$$

This, together with (11e), implies that

$$\begin{aligned} & E_{CH}(\phi^{n+1}) - E_{CH}(\phi^n) + \lambda \theta_{\eta} [(\eta^{n+1})^2 - (\eta^n)^2] \\ &\leq -\tau \eta^{n+1} \int_{\Omega} (\mathbf{u}^n \cdot \nabla \phi^n) \mu^{n+1} d\mathbf{x} \end{aligned} \quad (33)$$

On the other hand, taking the L^2 inner product of (11c) with $\tau \mathbf{u}^{n+1}$ gives us

$$\begin{aligned} & \int_{\Omega} (\mathbf{u}^{n+1} - \mathbf{u}^n + \tau q^{n+1} \mathbf{u}^n \cdot \nabla \mathbf{u}^n) \cdot \mathbf{u}^{n+1} d\mathbf{x} - \tau \eta^{n+1} \int_{\Omega} (\mu^{n+1} \nabla \phi^{n+1}) \cdot \mathbf{u}^{n+1} d\mathbf{x} \\ &= \tau \nu \int_{\Omega} \Delta \mathbf{u}^{n+1} \cdot \mathbf{u}^{n+1} d\mathbf{x} = -\tau \nu \|\nabla \mathbf{u}^{n+1}\|^2 \leq 0 \end{aligned} \quad (34)$$

where we have used the fact that $(\nabla p^{n+1}, \mathbf{u}^{n+1}) = 0$ due to the incompressibility condition (11d) and integration by parts. From Equations (11f) and (34), we obtain

$$\begin{aligned} & E_{NS}(\mathbf{u}^{n+1}) - E_{NS}(\mathbf{u}^n) + \theta_q [(q^{n+1})^2 - (q^n)^2] \\ &\leq \tau \eta^{n+1} \int_{\Omega} (\mathbf{u}^n \cdot \nabla \phi^n) \mu^{n+1} d\mathbf{x} \end{aligned} \quad (35)$$

The sum of (33) and (35) leads to

$$\begin{aligned} & E(\phi^{n+1}, \mathbf{u}^{n+1}) + \lambda \theta_{\eta} (\eta^{n+1})^2 + \theta_q (q^{n+1})^2 \\ &\leq E(\phi^n, \mathbf{u}^n) + \lambda \theta_{\eta} (\eta^n)^2 + \theta_q (q^n)^2 \end{aligned}$$

which completes the proof of Theorem 2. $\square$

Remark 1. From Theorem 2, we deduce that

$$\begin{aligned} & E(\phi^n, \mathbf{u}^n) + \lambda \theta_{\eta} (\eta^n)^2 + \theta_q (q^n)^2 \leq E(\phi^0, \mathbf{u}^0) + \lambda \theta_{\eta} (\eta^0)^2 + \theta_q (q^0)^2 \\ &= E(\phi^0, \mathbf{u}^0) + \lambda \theta_{\eta} + \theta_q \end{aligned} \quad (36)$$

for $0 \leq n \leq N$. Therefore, the discrete energy sequence $\{E(\phi^n, \mathbf{u}^n)\}_{n=0}^N$ is uniformly bounded.

Remark 2. Energy stability of the first-order DRLM scheme can also be proven for the CH-NS system with a forcing term $\mathbf{f}$ (cf. [15] and Section 5.5), that is, the momentum Equation (1c) is replaced by

$$\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} - \nu \Delta \mathbf{u} + \nabla p = \mu \nabla \phi + \mathbf{f} \quad (37)$$

In particular, the counterpart of (36) takes the following form

$$\begin{aligned} & E(\phi^n, \mathbf{u}^n) + \lambda \theta_{\eta} (\eta^n)^2 + \theta_q (q^n)^2 \\ &\leq e^{2T} \left(E(\phi^0, \mathbf{u}^0) + \lambda \theta_{\eta} + \theta_q + \frac{1}{2} \|\mathbf{f}\|_{L^\infty(0,T;L^2(\Omega))}^2 \right) \end{aligned} \quad (38)$$

provided that $\tau \leq \frac{1}{2}$ and $\mathbf{f} \in L^\infty(0,T;L^2(\Omega))$. Furthermore, under the no-flux/no-slip boundary conditions (3), the scheme is unconditionally energy stable if $\mathbf{f} \in L^\infty(0,T;H^{-1}(\Omega))$; that is, (36) becomes

$$\begin{aligned} & E(\phi^n, \mathbf{u}^n) + \lambda \theta_{\eta} (\eta^n)^2 + \theta_q (q^n)^2 \\ &\leq E(\phi^0, \mathbf{u}^0) + \lambda \theta_{\eta} + \theta_q + \frac{C_{\Omega}^2 T}{4\nu} \|\mathbf{f}\|_{L^\infty(0,T;H^{-1}(\Omega))}^2 \end{aligned} \quad (39)$$

for some constant $C_{\Omega} > 0$ only depending on Ω .

To prove (38), we first use similar arguments as in the proof of Theorem 2 to obtain

$$\begin{aligned} E(\phi^{n+1}, \mathbf{u}^{n+1}) + \lambda \theta_\eta (\eta^{n+1})^2 + \theta_q (q^{n+1})^2 \\ \leq E(\phi^n, \mathbf{u}^n) + \lambda \theta_\eta (\eta^n)^2 + \theta_q (q^n)^2 - \tau \nu \|\nabla \mathbf{u}^{n+1}\|^2 \\ + \tau \int_{\Omega} \mathbf{f}(t_{n+1}) \cdot \mathbf{u}^{n+1} d\mathbf{x} \end{aligned} \quad (40)$$

It is implied by Young's inequality and the definition of E in Equation (4) that

$$\begin{aligned} \tau \int_{\Omega} \mathbf{f}(t_{n+1}) \cdot \mathbf{u}^{n+1} d\mathbf{x} &\leq \frac{\tau}{2} \|\mathbf{f}(t_{n+1})\|^2 + \frac{\tau}{2} \|\mathbf{u}^{n+1}\|^2 \\ &\leq \frac{\tau}{2} \|\mathbf{f}(t_{n+1})\|^2 + \tau E(\phi^{n+1}, \mathbf{u}^{n+1}) \end{aligned} \quad (41)$$

Combining (40) and (41), we end up with

$$\begin{aligned} (1 - \tau) E(\phi^{n+1}, \mathbf{u}^{n+1}) + \lambda \theta_\eta (\eta^{n+1})^2 + \theta_q (q^{n+1})^2 \\ \leq E(\phi^n, \mathbf{u}^n) + \lambda \theta_\eta (\eta^n)^2 + \theta_q (q^n)^2 + \frac{\tau}{2} \|\mathbf{f}(t_{n+1})\|^2 \end{aligned}$$

or

$$\begin{aligned} E(\phi^{n+1}, \mathbf{u}^{n+1}) + \lambda \theta_\eta (\eta^{n+1})^2 + \theta_q (q^{n+1})^2 \\ \leq \frac{1}{1 - \tau} \left[E(\phi^n, \mathbf{u}^n) + \lambda \theta_\eta (\eta^n)^2 + \theta_q (q^n)^2 \right] + \frac{\tau}{2(1 - \tau)} \|\mathbf{f}(t_{n+1})\|^2 \end{aligned} \quad (42)$$

By repeatedly applying (42), one arrives at

$$\begin{aligned} E(\phi^n, \mathbf{u}^n) + \lambda \theta_\eta (\eta^n)^2 + \theta_q (q^n)^2 \\ \leq \frac{1}{(1 - \tau)^n} \left[E(\phi^0, \mathbf{u}^0) + \lambda \theta_\eta + \theta_q + \frac{\tau}{2} \sum_{i=0}^{n-1} (1 - \tau)^i \|\mathbf{f}(t_{i+1})\|^2 \right] \end{aligned}$$

This leads to (38), noting that $\frac{1}{(1 - \tau)^n} \leq (1 + 2\tau)^n \leq e^{2T}$ since $\tau \leq \frac{1}{2}$, and that $\tau \sum_{i=0}^{n-1} (1 - \tau)^i \leq 1$.

For the proof of (39), we use the Poincaré inequality for $\mathbf{u}^{n+1} \in \mathbf{H}_0^1(\Omega)$ (due to the no-flux/no-slip boundary conditions (3)):

$$\|\mathbf{u}^{n+1}\|_1 \leq C_\Omega \|\nabla \mathbf{u}^{n+1}\| \quad (43)$$

Therefore, the integral term on the right-hand side of (40) can be estimated as follows

$$\begin{aligned} \int_{\Omega} \mathbf{f}(t_{n+1}) \cdot \mathbf{u}^{n+1} d\mathbf{x} &\leq \|\mathbf{f}(t_{n+1})\|_{-1} \|\mathbf{u}^{n+1}\|_1 \\ &\leq C_\Omega \|\mathbf{f}(t_{n+1})\|_{-1} \|\nabla \mathbf{u}^{n+1}\| \leq \frac{C_\Omega^2 \tau}{4\nu} \|\mathbf{f}(t_{n+1})\|_{-1}^2 + \nu \|\nabla \mathbf{u}^{n+1}\|^2 \end{aligned} \quad (44)$$

From Equations (40) and (44), we obtain

$$\begin{aligned} E(\phi^{n+1}, \mathbf{u}^{n+1}) + \lambda \theta_\eta (\eta^{n+1})^2 + \theta_q (q^{n+1})^2 \\ \leq E(\phi^n, \mathbf{u}^n) + \lambda \theta_\eta (\eta^n)^2 + \theta_q (q^n)^2 + \frac{C_\Omega^2 \tau}{4\nu} \|\mathbf{f}(t_{n+1})\|_{-1}^2 \end{aligned}$$

Theorem 3. The second-order DRLM scheme (19) is unconditionally energy stable in the sense that for $1 \leq n \leq N - 1$, we have

$$\begin{aligned} \frac{3}{2} E(\phi^{n+1}, \mathbf{u}^{n+1}) - \frac{1}{2} E(\phi^n, \mathbf{u}^n) + \frac{\lambda}{2} \|\nabla(\phi^{n+1} - \phi^n)\|^2 \\ + \mathcal{E}_2(\eta^{n+1}, \eta^n, q^{n+1}, q^n) \\ \leq \frac{3}{2} E(\phi^n, \mathbf{u}^n) - \frac{1}{2} E(\phi^{n-1}, \mathbf{u}^{n-1}) + \frac{\lambda}{2} \|\nabla(\phi^n - \phi^{n-1})\|^2 \\ + \mathcal{E}_2(\eta^n, \eta^{n-1}, q^n, q^{n-1}) \end{aligned} \quad (45)$$

where

$$\begin{aligned} \mathcal{E}_2(\eta^{n+1}, \eta^n, q^{n+1}, q^n) &:= \lambda \theta_\eta \left[\frac{3}{2} (\eta^{n+1})^2 - \frac{1}{2} (\eta^n)^2 - 1 \right] \\ &\quad + \theta_q \left[\frac{3}{2} (q^{n+1})^2 - \frac{1}{2} (q^n)^2 - 1 \right] \end{aligned}$$

Proof. By taking the L^2 inner product of (19b) with $3\phi^{n+1} - 4\phi^n + \phi^{n-1}$ and (19a) with $2\tau\mu^{n+1}$, we obtain

$$\begin{aligned} \lambda \int_{\Omega} \left(-\Delta \phi^{n+1} + \eta^{n+1} F'(\tilde{\phi}^{n+1}) \right) (3\phi^{n+1} - 4\phi^n + \phi^{n-1}) d\mathbf{x} \\ = \int_{\Omega} \mu^{n+1} (3\phi^{n+1} - 4\phi^n + \phi^{n-1}) d\mathbf{x} \\ = -2\tau \eta^{n+1} \int_{\Omega} (\tilde{\mathbf{u}}^{n+1} \cdot \nabla \tilde{\phi}^{n+1}) \mu^{n+1} d\mathbf{x} - 2\tau M \|\nabla \mu^{n+1}\|^2 \\ \leq -2\tau \eta^{n+1} \int_{\Omega} (\tilde{\mathbf{u}}^{n+1} \cdot \nabla \tilde{\phi}^{n+1}) \mu^{n+1} d\mathbf{x} \end{aligned} \quad (46)$$

We observe from integration by parts that

$$\begin{aligned} \int_{\Omega} (-\Delta \phi^{n+1}) (3\phi^{n+1} - 4\phi^n + \phi^{n-1}) d\mathbf{x} \\ = \int_{\Omega} \nabla \phi^{n+1} \cdot \nabla (3\phi^{n+1} - 4\phi^n + \phi^{n-1}) d\mathbf{x} \\ \geq \frac{1}{2} (3\|\nabla \phi^{n+1}\|^2 - 4\|\nabla \phi^n\|^2 + \|\nabla \phi^{n-1}\|^2) \\ + \|\nabla(\phi^{n+1} - \phi^n)\|^2 - \|\nabla(\phi^n - \phi^{n-1})\|^2 \end{aligned} \quad (47)$$

In the last estimate, we have applied the following inequality

$$\begin{aligned} a(3a - 4b + c) &\geq \frac{1}{2} (3a^2 - 4b^2 + c^2) + (a - b)^2 - (b - c)^2, \\ &\quad \forall a, b, c \in \mathbb{R} \end{aligned}$$

Combining (46) and (47), we arrive at

$$\begin{aligned} \lambda \left[\frac{1}{2} (3\|\nabla \phi^{n+1}\|^2 - 4\|\nabla \phi^n\|^2 + \|\nabla \phi^{n-1}\|^2) + \|\nabla(\phi^{n+1} - \phi^n)\|^2 - \|\nabla(\phi^n - \phi^{n-1})\|^2 \right] \\ + \eta^{n+1} \int_{\Omega} F'(\tilde{\phi}^{n+1}) (3\phi^{n+1} - 4\phi^n + \phi^{n-1}) d\mathbf{x} \\ \leq -2\tau \eta^{n+1} \int_{\Omega} (\tilde{\mathbf{u}}^{n+1} \cdot \nabla \tilde{\phi}^{n+1}) \mu^{n+1} d\mathbf{x} \end{aligned}$$

This, together with (19e), leads to

$$\begin{aligned} 3E_{CH}(\phi^{n+1}) - 4E_{CH}(\phi^n) + E_{CH}(\phi^{n-1}) \\ + \lambda \|\nabla(\phi^{n+1} - \phi^n)\|^2 - \lambda \|\nabla(\phi^n - \phi^{n-1})\|^2 \\ + \lambda \theta_\eta [3(\eta^{n+1})^2 - 4(\eta^n)^2 + (\eta^{n-1})^2] \\ \leq -2\tau \eta^{n+1} \int_{\Omega} (\tilde{\mathbf{u}}^{n+1} \cdot \nabla \tilde{\phi}^{n+1}) \mu^{n+1} d\mathbf{x} \end{aligned} \quad (48)$$

Taking the L^2 inner product of (19c) with $2\tau \mathbf{u}^{n+1}$ yields

$$\begin{aligned} & \int_{\Omega} (3\mathbf{u}^{n+1} - 4\mathbf{u}^n + \mathbf{u}^{n-1} + 2\tau q^{n+1} \tilde{\mathbf{u}}^{n+1} \cdot \nabla \tilde{\mathbf{u}}^{n+1}) \cdot \mathbf{u}^{n+1} d\mathbf{x} \\ & - 2\tau \eta^{n+1} \int_{\Omega} (\mu^{n+1} \nabla \phi^{n+1}) \cdot \mathbf{u}^{n+1} d\mathbf{x} \\ & = -2\tau \nu \|\nabla \mathbf{u}^{n+1}\|^2 \leq 0 \end{aligned} \quad (49)$$

From Equations (19f) and (49), we deduce that

$$\begin{aligned} & 3E_{NS}(\mathbf{u}^{n+1}) - 4E_{NS}(\mathbf{u}^n) + E_{NS}(\mathbf{u}^{n-1}) \\ & + \theta_q [3(q^{n+1})^2 - 4(q^n)^2 + (q^{n-1})^2] \\ & \leq 2\tau \eta^{n+1} \int_{\Omega} (\tilde{\mathbf{u}}^{n+1} \cdot \nabla \tilde{\phi}^{n+1}) \mu^{n+1} d\mathbf{x} \end{aligned} \quad (50)$$

Adding (48) and (50) results in

$$\begin{aligned} & 3E(\phi^{n+1}, \mathbf{u}^{n+1}) - 4E(\phi^n, \mathbf{u}^n) + E(\phi^{n-1}, \mathbf{u}^{n-1}) \\ & + \lambda \|\nabla(\phi^{n+1} - \phi^n)\|^2 - \lambda \|\nabla(\phi^n - \phi^{n-1})\|^2 \\ & + \lambda \theta_{\eta} [3(\eta^{n+1})^2 - 4(\eta^n)^2 + (\eta^{n-1})^2] \\ & + \theta_q [3(q^{n+1})^2 - 4(q^n)^2 + (q^{n-1})^2] \leq 0 \end{aligned}$$

which concludes the proof of Theorem 3. $\square$

Remark 3. It follows from Theorem 3 that

$$\begin{aligned} & \frac{3}{2} E(\phi^n, \mathbf{u}^n) - \frac{1}{2} E(\phi^{n-1}, \mathbf{u}^{n-1}) + \frac{\lambda}{2} \|\nabla(\phi^n - \phi^{n-1})\|^2 \\ & + \lambda \theta_{\eta} \left[\frac{3}{2} (\eta^n)^2 - \frac{1}{2} (\eta^{n-1})^2 \right] + \theta_q \left[\frac{3}{2} (q^n)^2 - \frac{1}{2} (q^{n-1})^2 \right] \\ & \leq C^* \end{aligned} \quad (51)$$

for $1 \leq n \leq N$, where C^* is given by

$$\begin{aligned} C^* &:= \frac{3}{2} E(\phi^1, \mathbf{u}^1) - \frac{1}{2} E(\phi^0, \mathbf{u}^0) + \frac{\lambda}{2} \|\nabla(\phi^1 - \phi^0)\|^2 \\ &+ \lambda \theta_{\eta} \left[\frac{3}{2} (\eta^1)^2 - \frac{1}{2} \right] + \theta_q \left[\frac{3}{2} (q^1)^2 - \frac{1}{2} \right] \end{aligned} \quad (52)$$

For $0 \leq n \leq N$, we define

$$z^n := E(\phi^n, \mathbf{u}^n) + \lambda \theta_{\eta} (\eta^n)^2 + \theta_q (q^n)^2 \quad (53)$$

We obtain from Equation (51) that

$$z^n \leq \frac{1}{3} z^{n-1} + \frac{2}{3} C^*, \quad 1 \leq n \leq N$$

which implies that

$$z^n \leq C^* + \frac{1}{3^n} (z^0 - C^*), \quad 0 \leq n \leq N \quad (54)$$

Thus, $\{z^n\}_{n=0}^N$ and, consequently, $\{E(\phi^n, \mathbf{u}^n)\}_{n=0}^N$ are uniformly bounded.

Remark 4. For the CH-NS system with an extra forcing term $\mathbf{f}$ (cf. (37) in Remark 2), the energy stability result (54) of the second-order DRLM scheme becomes

$$z^n \leq \hat{C} + \frac{1}{3^n} (z^0 - \hat{C}), \quad 0 \leq n \leq N \quad (55)$$

provided that $\tau \leq \frac{4}{5}$ and $\mathbf{f} \in L^\infty(0, T; L^2(\Omega))$, where

$$\hat{C} := \frac{15}{14} e^{2T} \left[(3 - 2\tau) z^1 + \lambda \|\nabla(\phi^1 - \phi^0)\|^2 + \frac{1}{2} \|\mathbf{f}\|_{L^\infty(0, T; L^2(\Omega))}^2 \right]$$

Moreover, the second-order DRLM scheme is unconditionally energy stable with no-flux/no-slip boundary conditions if $\mathbf{f} \in L^\infty(0, T; H^{-1}(\Omega))$:

$$z^n \leq \tilde{C} + \frac{1}{3^n} (z^0 - \tilde{C}), \quad 0 \leq n \leq N \quad (56)$$

where $\tilde{C} := C^* + \frac{C_\Omega^2 T}{4\nu} \|\mathbf{f}\|_{L^\infty(0, T; H^{-1}(\Omega))}^2$, C^* is given in Equation (52), and C_Ω denotes the Poincaré constant defined in Equation (43).

The proof of (56) is done in the same manner as in Remarks 2 and 3 and thus omitted. To derive (55), we follow the proof of Theorem 3 and obtain

$$\begin{aligned} & 3E(\phi^{n+1}, \mathbf{u}^{n+1}) - 4E(\phi^n, \mathbf{u}^n) + E(\phi^{n-1}, \mathbf{u}^{n-1}) + \lambda \|\nabla(\phi^{n+1} - \phi^n)\|^2 \\ & - \lambda \|\nabla(\phi^n - \phi^{n-1})\|^2 + \lambda \theta_{\eta} [3(\eta^{n+1})^2 - 4(\eta^n)^2 + (\eta^{n-1})^2] \\ & + \theta_q [3(q^{n+1})^2 - 4(q^n)^2 + (q^{n-1})^2] \\ & \leq -2\tau \nu \|\nabla \mathbf{u}^{n+1}\|^2 + 2\tau \int_{\Omega} \mathbf{f}(t_{n+1}) \cdot \mathbf{u}^{n+1} d\mathbf{x} \end{aligned} \quad (57)$$

Recalling the definition of $\{z^n\}$ in Equation (53) and Young's inequality (41), we deduce from Equation (57) that

$$\begin{aligned} & (3 - 2\tau) z^{n+1} - 4z^n + z^{n-1} + \lambda \|\nabla(\phi^{n+1} - \phi^n)\|^2 \\ & - \lambda \|\nabla(\phi^n - \phi^{n-1})\|^2 \leq \tau \|\mathbf{f}(t_{n+1})\|^2 \end{aligned}$$

which implies

$$\begin{aligned} & (3 - 2\tau) z^{n+1} - (2 - \sqrt{1 + 2\tau}) z^n + \lambda \|\nabla(\phi^{n+1} - \phi^n)\|^2 \\ & \leq \frac{2 + \sqrt{1 + 2\tau}}{3 - 2\tau} \left[(3 - 2\tau) z^n - (2 - \sqrt{1 + 2\tau}) z^{n-1} \right. \\ & \quad \left. + \lambda \|\nabla(\phi^n - \phi^{n-1})\|^2 \right] + \tau \|\mathbf{f}(t_{n+1})\|^2 \\ & \leq (1 + 2\tau) \left[(3 - 2\tau) z^n - (2 - \sqrt{1 + 2\tau}) z^{n-1} \right. \\ & \quad \left. + \lambda \|\nabla(\phi^n - \phi^{n-1})\|^2 \right] + \tau \|\mathbf{f}(t_{n+1})\|^2 \end{aligned} \quad (58)$$

Note that in the last inequality, we have used the fact that $\tau \leq \frac{4}{5}$. Repeatedly applying the estimate (58) yields

$$\begin{aligned} & (3 - 2\tau) z^{n+1} - (2 - \sqrt{1 + 2\tau}) z^n + \lambda \|\nabla(\phi^{n+1} - \phi^n)\|^2 \\ & \leq (1 + 2\tau)^n \left[(3 - 2\tau) z^1 - (2 - \sqrt{1 + 2\tau}) z^0 + \lambda \|\nabla(\phi^1 - \phi^0)\|^2 \right. \\ & \quad \left. + \tau \sum_{i=1}^n (1 + 2\tau)^{-i} \|\mathbf{f}(t_{i+1})\|^2 \right] \\ & \leq e^{2T} \left[(3 - 2\tau) z^1 + \lambda \|\nabla(\phi^1 - \phi^0)\|^2 + \frac{1}{2} \|\mathbf{f}\|_{L^\infty(0, T; L^2(\Omega))}^2 \right] =: \hat{C}_0 \end{aligned} \quad (59)$$

By dropping the non-negative term $\lambda \|\nabla(\phi^{n+1} - \phi^n)\|^2$ on the left-hand side of (59), we obtain

$$\begin{aligned} z^{n+1} &\leq \frac{2 - \sqrt{1 + 2\tau}}{3 - 2\tau} z^n + \frac{\hat{C}_0}{3 - 2\tau} \\ &= \frac{1}{2 + \sqrt{1 + 2\tau}} z^n + \frac{\hat{C}_0}{3 - 2\tau} \leq \frac{1}{3} z^n + \frac{\hat{C}_0}{3 - 2\tau}, \quad n \geq 0 \end{aligned} \quad (60)$$

Since $\tau \leq \frac{4}{5}$, we have $3 - 2\tau \geq \frac{7}{5}$. Hence, we deduce from Equation (60) that

$$z^{n+1} \leq \frac{1}{3} z^n + \frac{5}{7} \hat{C}_0, \quad n \geq 0$$

With $\hat{C} = \frac{15}{14} \hat{C}_0$, the estimate (55) is thus proved.

Remark 5. Convergence analysis of the proposed DRLM schemes (cf. (11) and (19)) for the coupled CH-NS system—and even for the incompressible Navier-Stokes equations alone [37]—remains an open question and is part of our current research. The main challenges arise from (i) the explicit treatment of nonlinear terms $F'(\phi)$ and $\mathbf{u} \cdot \nabla \mathbf{u}$, (ii) the complicated coupling between primary variables $(\phi, \mu, \mathbf{u}, p)$ and scalar Lagrange multipliers (η, q) , and (iii) the presence of quadratic terms involving the Lagrange multipliers, such as $(\eta^{n+1})^2$ and $(q^{n+1})^2$. We note that for numerical schemes where the nonlinear terms are treated fully implicitly or semi-implicitly, error analysis can be carried out as in References [19, 42]. For SAV-based schemes with explicit treatment of nonlinear terms [43–45], although energy stability is obtained with respect to a modified energy, certain error estimates can be derived due to the linear structure of auxiliary variables. We refer to [46] for the analysis of a fully discrete SAV scheme that incorporates both a semi-implicit treatment of the nonlinear convection term $\mathbf{u} \cdot \nabla \mathbf{u}$ and an explicit treatment of $F'(\phi)$ via a suitable auxiliary variable.

5 | Numerical Experiments

We carry out various numerical experiments to illustrate the performance of the two proposed DRLM schemes (11) and (19). In the first test case, a manufactured solution is considered to verify the temporal convergence of both first- and second-order DRLM schemes and to compare their performance with the SAV-based schemes. We also investigate the role of the regularization parameters in DRLM schemes to ensure the accuracy of the Lagrange multipliers while allowing for large time step sizes (cf. Tables 5 and 6). The mass conservation and energy stability of the DRLM schemes are confirmed in the second test case. We then apply the second-order DRLM scheme for more realistic test cases, including shape relaxation, spinodal decomposition, and dripping droplet in two and three dimensions, to demonstrate its robustness and ability to capture the interface dynamics of two-phase systems.

For spatial discretization, different methods such as finite difference, finite element, or finite volume methods can be applied. For simplicity, we consider the Marker-And-Cell (MAC) discretization on a staggered grid, where scalar variables such as

the phase variable, chemical potential, and pressure are located at the cell centers, while the velocity components are positioned at the cell faces. We refer to [26, 37] for the details on MAC discretization. In the following, we consider no-flux boundary conditions for the phase variable and chemical potential, and no-slip conditions for the velocity (cf. (3)); similar results can be obtained with periodic boundary conditions. The spatial mesh sizes are uniform, $h = h_x = h_y$ in 2D and $h = h_x = h_y = h_z$ in 3D.

Note that the two DRLM schemes involve solving the following type of fourth-order linear equation (cf. (13, 14) and (20, 21)):

$$(\alpha I + \lambda M \Delta^2) \phi = f \quad (61)$$

and the generalized Stokes system (cf. (16, 17) and (23, 24))

$$\alpha \mathbf{u} - \nu \Delta \mathbf{u} + \nabla p = \mathbf{g}, \quad \nabla \cdot \mathbf{u} = 0 \quad (62)$$

where f and $\mathbf{g}$ are given functions, $\alpha = \frac{1}{\tau}$ for the first-order DRLM scheme (11) and $\alpha = \frac{3}{2\tau}$ for the second-order counterpart (19). With the MAC spatial discretization, (61) can be solved via the Discrete Fourier Transform (for periodic boundary conditions) and the Discrete Cosine Transform (for homogeneous Neumann boundary conditions (3)); we refer to [47] and [48] for an overview of Fast Fourier Transform-based algorithms for the Allen-Cahn and Cahn-Hilliard equations. On the other hand, the Stokes system (62) can be transformed into a saddle point problem, where various effective preconditioners [13, 37, 49] are available to enhance the performance of iterative solvers such as GMRES.

For comparison purposes, we consider the multiple scalar auxiliary variables (MSAV) approach introduced in Reference [27], which is briefly summarized as follows. Let $\kappa > 0$ be a stabilization constant, and define

$$G(\phi) := F(\phi) - \frac{\kappa}{2} \phi^2, \quad \text{and} \quad E_1(\phi) = \int_{\Omega} G(\phi) dx$$

The two extra space-independent variables are defined by

$$r(t) = \sqrt{E_1(\phi) + \delta}, \quad s(t) = e^{-t/T} \quad (63)$$

for some constant $\delta > 0$ to ensure that $E_1(\phi) + \delta > 0$. In light of (63), the original CH-NS system (1) is reformulated into the following equivalent form

$$\frac{\partial \phi}{\partial t} + \frac{r}{\sqrt{E_1(\phi) + \delta}} \mathbf{u} \cdot \nabla \phi = M \Delta \mu, \quad \text{in } \Omega \times (0, T] \quad (64a)$$

$$\mu = \lambda \left(-\Delta \phi + \kappa \phi + \frac{r}{\sqrt{E_1(\phi) + \delta}} G'(\phi) \right), \quad \text{in } \Omega \times (0, T] \quad (64b)$$

$$\frac{\partial \mathbf{u}}{\partial t} + e^{t/T} s(t) \mathbf{u} \cdot \nabla \mathbf{u} - \nu \Delta \mathbf{u} + \nabla p = \frac{r}{\sqrt{E_1(\phi) + \delta}} \mu \nabla \phi, \quad \text{in } \Omega \times (0, T] \quad (64c)$$

$$\nabla \cdot \mathbf{u} = 0, \quad \text{in } \Omega \times (0, T] \quad (64d)$$

$$\frac{dr}{dt} = \frac{1}{2\sqrt{E_1(\phi) + \delta}} \left[\int_{\Omega} G'(\phi) \frac{\partial \phi}{\partial t} d\mathbf{x} + \frac{1}{\lambda} \int_{\Omega} ((\mathbf{u} \cdot \nabla \phi) \mu - (\mu \nabla \phi) \cdot \mathbf{u}) d\mathbf{x} \right], \quad \text{in } (0, T] \quad (64e)$$

$$\frac{ds}{dt} = -\frac{1}{T} s + e^{t/T} \int_{\Omega} (\mathbf{u} \cdot \nabla \mathbf{u}) \cdot \mathbf{u} d\mathbf{x}, \quad \text{in } (0, T] \quad (64f)$$

with the initial conditions for the scalar variables: $r(0) = \sqrt{E_1(\phi|_{t=0}) + \delta}$ and $s(0) = 1$. From Equation (64), the first- and second-order MSAV schemes can be derived based on the associated BDF methods, as detailed in Reference [27].

5.1 | Convergence Test

By adding suitable forcing terms to the right-hand sides of (1a) and (1c), we suppose that the exact solution of the CH-NS Equations (1) is given by:

$$\phi(x, y, t) = \cos(\pi x) \cos(\pi y) \sin(t),$$

$$p(x, y, t) = \cos(\pi x) \sin(\pi y) \sin(t),$$

$$\mathbf{u}(x, y, t) = [\pi \sin^2(\pi x) \sin(2\pi y) \sin(t), -\pi \sin(2\pi x) \sin^2(\pi y) \sin(t)]^T \quad (65)$$

We take the domain $\Omega = (0, 1)^2$, $\varepsilon = 0.1$, $M = 1$, $\lambda = 0.001$, $\nu = 0.1$, $\theta_\eta = 100/\varepsilon^2$, $\theta_q = 100$ and vary the time step size and spatial mesh size such that $h = \frac{\sqrt{\tau}}{4}$ for the first-order DRLM scheme and $h = \frac{\tau}{2}$ for the second-order one, where $\tau \in \left\{ \frac{1}{16}, \frac{1}{32}, \frac{1}{64}, \frac{1}{128}, \frac{1}{256} \right\}$. The L^2 and L^∞ errors of all variables, along with the absolute errors of the Lagrange multipliers at the final time $T = 1$, are reported in Table 1 for the first-order DRLM scheme and in Table 2 for the second-order counterpart. The results confirm the expected order of convergence for all quantities. We note that, in contrast to other approaches that employ projection methods [21, 27, 30], which often result in a lower convergence order for pressure due to artificial boundary conditions, our DRLM schemes maintain consistent convergence for pressure and other variables.

Next, we compare DRLM schemes with the MSAV schemes [27] derived from the reformulated system (64); for consistency, the velocity and pressure are coupled via a Stokes system rather than

TABLE 1 | Errors and convergence rates of the CH-NS variables and Lagrange multipliers by the first-order DRLM scheme (11) with $h = \frac{\sqrt{\tau}}{4}$.

τ	$\ e_\phi\ _2$	$\ e_\phi\ _\infty$	$\ e_\mu\ _2$	$\ e_\mu\ _\infty$	$ \eta^N - 1 $
1/16	1.45e-01	2.30e-01	7.05e-03	1.71e-02	3.56e-05
1/32	5.27e-02 [1.46]	8.31e-02 [1.47]	2.31e-03 [1.61]	5.54e-03 [1.63]	1.16e-05 [1.62]
1/64	2.25e-02 [1.23]	3.84e-02 [1.11]	9.17e-04 [1.33]	2.02e-03 [1.46]	4.95e-06 [1.23]
1/128	1.06e-02 [1.09]	1.87e-02 [1.04]	4.12e-04 [1.15]	8.62e-04 [1.23]	2.37e-06 [1.06]
1/256	5.21e-03 [1.02]	9.21e-03 [1.02]	1.96e-04 [1.07]	4.08e-04 [1.08]	1.17e-06 [1.02]
τ	$\ e_u\ _2$	$\ e_u\ _\infty$	$\ e_p\ _2$	$\ e_p\ _\infty$	$ q^N - 1 $
1/16	8.95e-03	1.51e-02	2.95e-01	9.50e-01	4.28e-04
1/32	4.41e-03 [1.02]	7.53e-03 [1.00]	1.47e-01 [1.00]	4.89e-01 [0.96]	2.12e-04 [1.01]
1/64	2.38e-03 [0.89]	4.08e-03 [0.88]	7.41e-02 [0.99]	2.44e-01 [1.00]	1.06e-04 [1.00]
1/128	1.22e-03 [0.96]	2.10e-03 [0.96]	3.71e-02 [1.00]	1.23e-01 [0.99]	5.27e-05 [1.01]
1/256	6.00e-04 [1.02]	1.03e-03 [1.03]	1.85e-02 [1.00]	6.13e-02 [1.00]	2.63e-05 [1.00]

TABLE 2 | Errors and convergence rates of the CH-NS variables and Lagrange multipliers by the second-order DRLM scheme (19) with $h = \frac{\tau}{2}$.

τ	$\ e_\phi\ _2$	$\ e_\phi\ _\infty$	$\ e_\mu\ _2$	$\ e_\mu\ _\infty$	$ \eta^N - 1 $
1/16	3.93e-03	9.23e-03	3.84e-04	7.79e-04	4.27e-06
1/32	9.54e-04 [2.04]	2.09e-03 [2.14]	6.47e-05 [2.57]	1.36e-04 [2.52]	1.14e-06 [1.91]
1/64	2.67e-04 [1.84]	5.98e-04 [1.81]	1.69e-05 [1.94]	3.81e-05 [1.84]	2.95e-07 [1.95]
1/128	6.98e-05 [1.94]	1.58e-04 [1.92]	4.31e-06 [1.97]	9.95e-06 [1.94]	7.53e-08 [1.97]
1/256	1.78e-05 [1.97]	4.06e-05 [1.96]	1.09e-06 [1.98]	2.54e-06 [1.97]	1.90e-08 [1.99]
τ	$\ e_u\ _2$	$\ e_u\ _\infty$	$\ e_p\ _2$	$\ e_p\ _\infty$	$ q^N - 1 $
1/16	4.05e-03	6.35e-03	4.13e-03	9.07e-03	3.11e-05
1/32	1.02e-03 [1.99]	1.59e-03 [2.00]	1.19e-03 [1.80]	2.56e-03 [1.82]	7.46e-06 [2.06]
1/64	2.55e-04 [2.00]	3.99e-04 [1.99]	3.14e-04 [1.92]	6.70e-04 [1.93]	1.83e-06 [2.03]
1/128	6.38e-05 [2.00]	9.99e-05 [2.00]	8.04e-05 [1.97]	1.71e-04 [1.97]	4.52e-07 [2.02]
1/256	1.59e-05 [2.00]	2.50e-05 [2.00]	2.03e-05 [1.99]	4.32e-05 [1.98]	1.12e-07 [2.01]

the original pressure-correction approach [27]. The parameters ε , M , λ , ν , T , as well as the mesh size h and time step size τ , are chosen as before, while the stabilization constant κ and the positive constant δ are given by $\kappa = 5/\varepsilon^2$ and $\delta = 35/(4\varepsilon^2)$ as in Reference [27]. Errors by the first- and second-order MSAV schemes are reported in Tables 3 and 4, respectively; expected order of convergence is confirmed for all quantities. We observe that DRLM schemes provide significantly improved accuracy for the CH-related variables compared to MSAV counterparts, as DRLM schemes require no stabilization. Both DRLM and MSAV schemes provide comparable results for the NS-related variables.

To investigate the effect of the regularization parameters on the proposed DRLM schemes, we vary θ_η and θ_q and compute absolute errors of the Lagrange multipliers at the final time $T = 1$. These errors are presented in Tables 5 and 6 for the first- and second-order DRLM schemes, respectively. We observe that as θ_η and θ_q increase, the errors in both Lagrange multipliers become much smaller, resulting in better accuracy of the numerical solutions, especially with large time step sizes. Hence, we will choose

sufficiently large values for θ_η and θ_q in the subsequent numerical experiments.

5.2 | Mass Conservation and Energy Dissipation Test

Next, we test the properties of mass conservation and energy dissipation by the proposed DRLM schemes. Let us consider the CH-NS system (1) in the domain $\Omega = (0, 1)^2$ with the following initial conditions [21]:

$$\begin{aligned}\phi_0(x, y) &= 0.24 \cos(2\pi x) \cos(2\pi y) + 0.4 \cos(\pi x) \cos(3\pi y), \\ \mathbf{u}_0(x, y) &= [-\sin^2(\pi x) \sin(2\pi y), \sin(2\pi x) \sin^2(\pi y)]^T\end{aligned}\quad (66)$$

The parameters are set as follows:

$$\begin{aligned}\varepsilon &= 0.1, \quad M = 1, \quad \lambda = 0.01, \quad \nu = 0.01, \\ \theta_\eta &= 1000/\varepsilon^2, \quad \theta_q = 100\end{aligned}$$

TABLE 3 | Errors and convergence rates of the CH-NS variables and scalar auxiliary variables by the first-order MSAV scheme with $h = \frac{\sqrt{\tau}}{4}$.

τ	$\ e_\phi\ _2$	$\ e_\phi\ _\infty$	$\ e_\mu\ _2$	$\ e_\mu\ _\infty$	$ e_r _\infty$
1/16	2.40e-01	4.59e-01	2.91e-02	4.65e-02	5.94e-02
1/32	1.51e-01 [0.67]	3.14e-01 [0.55]	1.50e-02 [0.96]	2.76e-02 [0.75]	1.64e-01 [-1.47]
1/64	7.77e-02 [0.96]	1.69e-01 [0.89]	6.08e-03 [1.30]	1.28e-02 [1.11]	1.31e-01 [0.32]
1/128	3.57e-02 [1.12]	8.16e-02 [1.05]	2.29e-03 [1.41]	5.13e-03 [1.32]	6.64e-02 [0.98]
1/256	1.66e-02 [1.10]	3.91e-02 [1.06]	9.74e-04 [1.23]	2.16e-03 [1.25]	3.22e-02 [1.04]
τ	$\ e_u\ _2$	$\ e_u\ _\infty$	$\ e_p\ _2$	$\ e_p\ _\infty$	$ e_s _\infty$
1/16	1.02e-02	1.70e-02	2.17e-01	7.20e-01	1.00e-02
1/32	4.17e-03 [1.29]	6.90e-03 [1.30]	1.04e-01 [1.06]	3.58e-01 [1.01]	5.42e-03 [0.88]
1/64	2.05e-03 [1.02]	3.25e-03 [1.09]	5.29e-02 [0.98]	1.80e-01 [0.99]	2.80e-03 [0.95]
1/128	1.07e-03 [0.94]	1.68e-03 [0.95]	2.66e-02 [0.99]	9.08e-02 [0.99]	1.42e-03 [0.98]
1/256	5.45e-04 [0.97]	8.51e-04 [0.98]	1.32e-02 [1.01]	4.49e-02 [1.02]	7.14e-04 [0.99]

TABLE 4 | Errors and convergence rates of the CH-NS variables and scalar auxiliary variables by the second-order MSAV scheme with $h = \frac{\tau}{2}$.

τ	$\ e_\phi\ _2$	$\ e_\phi\ _\infty$	$\ e_\mu\ _2$	$\ e_\mu\ _\infty$	$ e_r _\infty$
1/16	7.54e-02	1.35e-01	7.93e-03	1.69e-02	2.14e-01
1/32	1.05e-02 [2.84]	2.16e-02 [2.64]	7.16e-04 [3.47]	1.43e-03 [3.56]	2.59e-02 [3.05]
1/64	2.25e-03 [2.22]	4.73e-03 [2.19]	1.20e-04 [2.58]	2.62e-04 [2.45]	6.91e-03 [1.91]
1/128	5.78e-04 [1.96]	1.19e-03 [1.99]	2.99e-05 [2.00]	6.76e-05 [1.95]	1.78e-03 [1.96]
1/256	1.47e-04 [1.98]	3.01e-04 [1.98]	7.53e-06 [1.99]	1.72e-05 [1.97]	4.50e-04 [1.98]
τ	$\ e_u\ _2$	$\ e_u\ _\infty$	$\ e_p\ _2$	$\ e_p\ _\infty$	$ e_s _\infty$
1/16	2.92e-03	4.39e-03	4.92e-03	9.45e-03	6.91e-04
1/32	9.32e-04 [1.65]	1.41e-03 [1.64]	1.67e-03 [1.56]	3.86e-03 [1.29]	1.56e-04 [2.15]
1/64	2.43e-04 [1.94]	3.72e-04 [1.92]	4.96e-04 [1.75]	1.18e-03 [1.71]	3.80e-05 [2.04]
1/128	6.11e-05 [1.99]	9.35e-05 [1.99]	1.27e-04 [1.97]	3.05e-04 [1.95]	9.41e-06 [2.01]
1/256	1.53e-05 [2.00]	2.34e-05 [2.00]	3.20e-05 [1.99]	7.69e-05 [1.99]	2.35e-06 [2.00]

TABLE 5 | Errors of the Lagrange multipliers by the first-order DRLM scheme (11) with $h = \frac{\sqrt{\tau}}{4}$ and different regularization parameters.

τ	$\theta_\eta = 0.1/\varepsilon^2, \theta_q = 0.1$		$\theta_\eta = 1/\varepsilon^2, \theta_q = 1$		$\theta_\eta = 10/\varepsilon^2, \theta_q = 10$	
	$ \eta^N - 1 $	$ q^N - 1 $	$ \eta^N - 1 $	$ q^N - 1 $	$ \eta^N - 1 $	$ q^N - 1 $
1/16	2.77e-02	3.65e-01	3.46e-03	4.20e-02	3.55e-04	4.27e-03
1/32	8.54e-03	1.94e-01	1.13e-03	2.10e-02	1.16e-04	2.12e-03
1/64	3.63e-03	1.01e-01	4.80e-04	1.05e-02	4.93e-05	1.05e-03
1/128	1.75e-03	5.14e-02	2.30e-04	5.25e-03	2.37e-05	5.27e-04
1/256	8.71e-04	2.60e-02	1.14e-04	2.63e-03	1.17e-05	2.63e-04

TABLE 6 | Errors of the Lagrange multipliers by the second-order DRLM scheme (19) with $h = \frac{\tau}{2}$ and different regularization parameters.

τ	$\theta_\eta = 0.1/\varepsilon^2, \theta_q = 0.1$		$\theta_\eta = 1/\varepsilon^2, \theta_q = 1$		$\theta_\eta = 10/\varepsilon^2, \theta_q = 10$	
	$ \eta^N - 1 $	$ q^N - 1 $	$ \eta^N - 1 $	$ q^N - 1 $	$ \eta^N - 1 $	$ q^N - 1 $
1/16	3.22e-03	3.09e-02	4.15e-04	3.11e-03	4.26e-05	3.11e-04
1/32	8.41e-04	7.45e-03	1.10e-04	7.46e-04	1.13e-05	7.46e-05
1/64	2.18e-04	1.83e-03	2.86e-05	1.83e-04	2.94e-06	1.83e-05
1/128	5.54e-05	4.52e-04	7.30e-06	4.52e-05	7.51e-07	4.52e-06
1/256	1.40e-05	1.12e-04	1.84e-06	1.12e-05	1.90e-07	1.12e-06

The evolution of mass, original energy, and error functions $\mathcal{E}_1$ and $\mathcal{E}_2$ associated with the Lagrange multipliers (cf. Theorems 2 and 3) up to $T = 10$ is shown in Figure 1 for both DRLM schemes with different spatial mesh sizes and time step sizes, where $h = \tau \in \{4 \times 10^{-3}, 2 \times 10^{-3}\}$. We clearly observe the conservation of mass and the dissipation of energy by the numerical solutions over time. Additionally, the error functions $\mathcal{E}_1$ and $\mathcal{E}_2$ in the energy stability results (30) and (45) are negligible compared to the energy functional, with their magnitudes getting closer to zero as h and τ are refined.

5.3 | Shape Relaxation

We now simulate the evolution of a square-shaped fluid bubble in the domain $\Omega = (0, 1)^2$. The phase variable is initialized with a value of 1 inside the square bubble $[0.25, 0.75]^2$ and -1 outside. The initial velocity is set to be zero, and other parameters are given by

$$\varepsilon \in \{0.01, 0.02\}, \quad M = 0.002, \quad \lambda = 0.01, \quad \nu = 1, \\ \theta_\eta = 1000/\varepsilon^2, \quad \theta_q = 100$$

We perform the simulation until $T = 1$ using the second-order DRLM scheme with $h = 4 \times 10^{-3}$, $\tau = 2 \times 10^{-3}$ for $\varepsilon = 0.02$, and $\tau = 2 \times 10^{-4}$ for $\varepsilon = 0.01$. The evolution of the phase variable at different times for $\varepsilon \in \{0.01, 0.02\}$ is shown in Figure 2, where the dashed circles, centered at $(0.5, 0.5)$ with radius $\frac{1}{2\sqrt{\pi}}$, indicate the interface at steady state. Under the effect of surface tension, the initial square shape eventually transitions into a circle that closely matches the dashed one, with the circle forming at an earlier stage for $\varepsilon = 0.01$ than for $\varepsilon = 0.02$. Our second-order DRLM scheme accurately captures this dynamic within a short time interval. Figure 3 confirms the energy decay property of

the numerical solutions, with the total energy stabilizing at approximately $t = 1$ for $\varepsilon = 0.02$ and near $t = 0.6$ for $\varepsilon = 0.01$. In contrast, existing stabilized schemes (e.g., [25, 26]) allow for relatively larger time step sizes but require significantly longer times for the shape relaxation process due to artificial dissipation introduced by stabilization constants.

5.4 | Spinodal Decomposition

The CH-NS system (1) can serve as a model for spinodal decomposition in a binary fluid mixture [21, 30, 50, 51]. This decomposition begins with a small perturbation to the initial concentration field, set as $\phi_0(x, y) = \bar{\phi}_0 + 0.001 \text{rand}(x, y)$, where $\text{rand}(x, y)$ denotes a random number in $[-1, 1]$ following the normal distribution, and the volume fraction $\bar{\phi}_0 \in \{0, 0.5\}$. The initial velocity is set to zero, that is, $\mathbf{u}_0 = \mathbf{0}$. We take the domain $\Omega = (0, 1)^2$ with the following parameters:

$$\varepsilon = 0.01, \quad M = 0.001, \quad \lambda = 0.01, \quad \nu = 1, \\ \theta_\eta = 1000/\varepsilon^2, \quad \theta_q = 1000$$

We perform the simulation until $T = 25$ for $\bar{\phi}_0 = 0$ and $T = 50$ for $\bar{\phi}_0 = 0.5$ using the second-order DRLM scheme with $h = 4 \times 10^{-3}$ and $\tau = 4 \times 10^{-4}$. Snapshots of the phase variable at different times are shown in Figure 4 for $\bar{\phi}_0 = 0$ and in Figure 5 for $\bar{\phi}_0 = 0.5$. Both simulations exhibit the phase separation process, with the latter (i.e., $\bar{\phi}_0 = 0.5$) requiring a longer time to reach the equilibrium state. At equilibrium, a pipe-like phase structure is observed in the case of $\bar{\phi}_0 = 0$ while a circular phase structure is observed in the case of $\bar{\phi}_0 = 0.5$. Figure 6 displays the original energy evolution for each case, where the energy functional decreases significantly at the early state before gradually stabilizing as it approaches the final time.

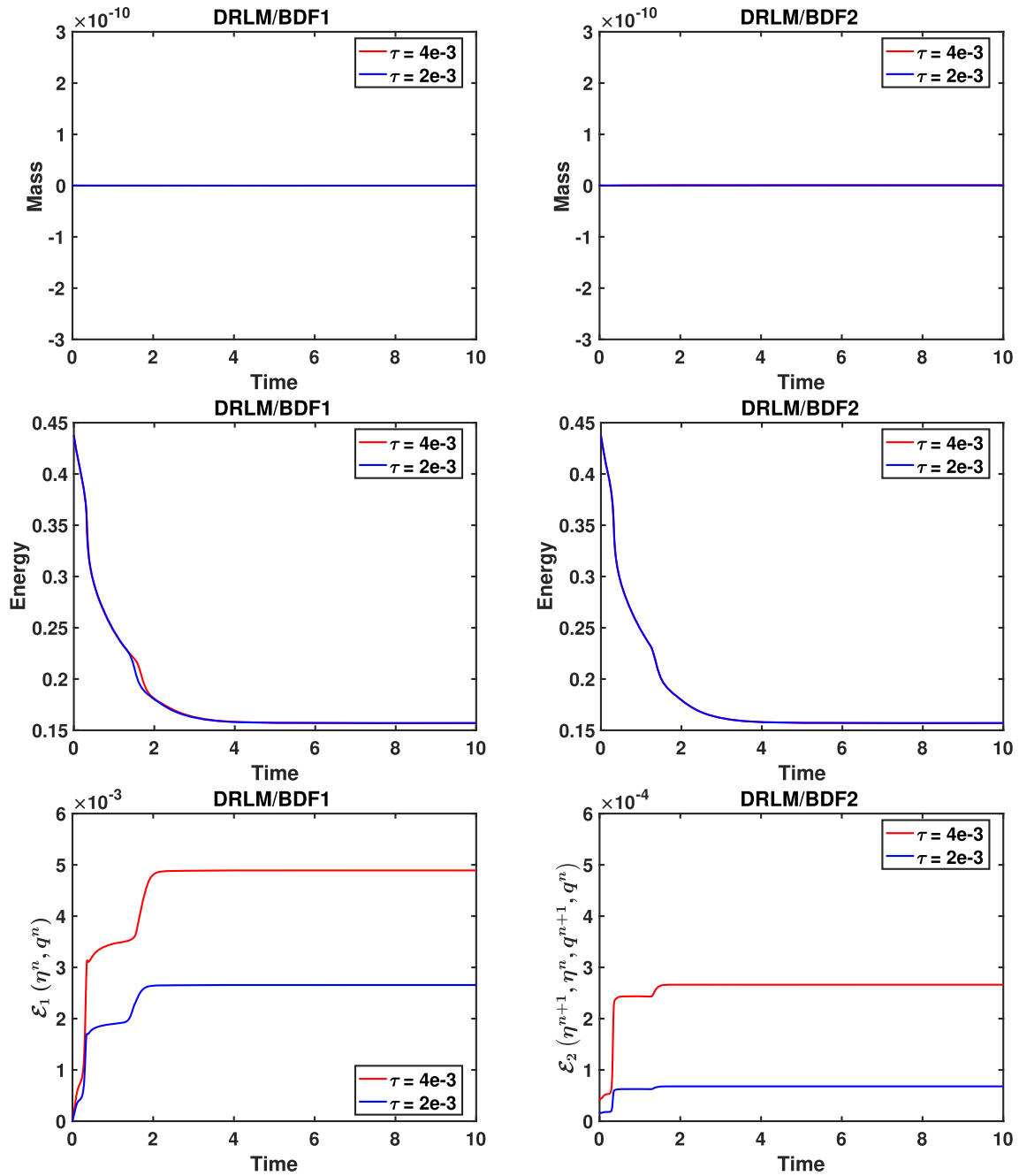


FIGURE 1 | Evolution of mass, original energy, and error functions $\mathcal{E}_i$ ($i = 1, 2$) produced by the DRLM schemes with $h = \tau \in \{4 \times 10^{-3}, 2 \times 10^{-3}\}$. Left column: The first-order scheme (11); Right column: The second-order scheme (19).

5.5 | Dripping Droplet

We finally perform two-dimensional and three-dimensional simulations of the dripping dynamics of a droplet falling from the top wall under the gravitational force [28, 30]. By assuming the density difference of the liquid drop and the ambient fluid is small, we reformulate the momentum Equation (1c) using Boussinesq approximations [52–54] as follows:

$$\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} - \nu \Delta \mathbf{u} + \nabla p = \mu \nabla \phi + \chi \rho(\phi) \mathbf{g} \quad (67)$$

where $\chi \rho(\phi) \mathbf{g}$ is a buoyancy term with $\chi \rho(\phi) = \chi(\phi - \bar{\phi})$, $\mathbf{g} = [0, -1]^T$ in 2D and $\mathbf{g} = [0, 0, -1]^T$ in 3D.

For the two-dimensional case, we set the computational domain $\Omega = (0, 1) \times (0, 2)$, the initial conditions:

$$\begin{aligned} \phi_0(x, y) &= \tanh \left(\frac{0.32 - \sqrt{(x - 0.5)^2 + (y - 2.1)^2}}{\varepsilon} \right), \\ \mathbf{u}_0(x, y) &= \mathbf{0} \end{aligned} \quad (68)$$

and the parameters:

$$\begin{aligned} \varepsilon &= 0.01, \quad M = 0.01, \quad \lambda = 0.001, \quad \theta_\eta = 1000/\varepsilon^2, \\ \theta_q &= 1000, \quad \chi = 10, \quad \bar{\phi} = 0 \end{aligned}$$

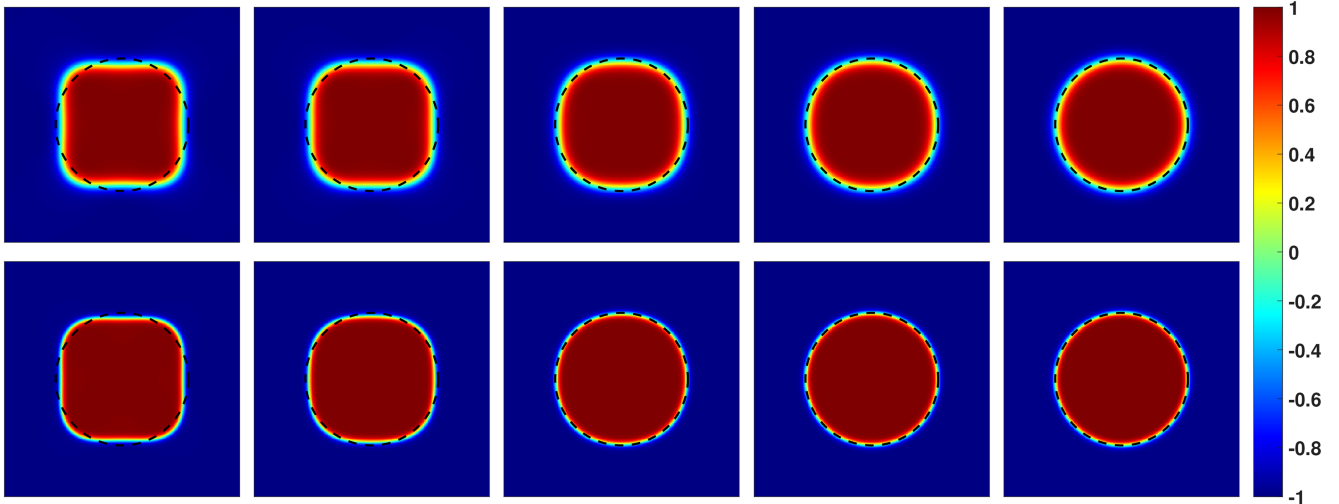


FIGURE 2 | Shape relaxation of surface tension-driven flow at $t = 0.1, 0.2, 0.4, 0.6$, and 1 (from left to right) by the second-order DRLM scheme (19) with $\varepsilon = 0.02$ (top) and $\varepsilon = 0.01$ (bottom).

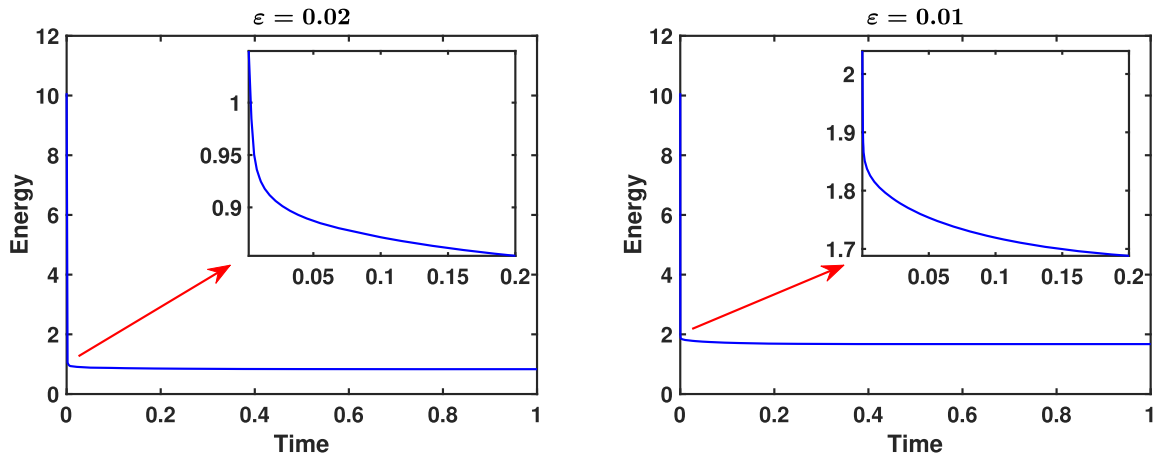


FIGURE 3 | Evolution of the original energy during the shape relaxation by the second-order DRLM scheme (19) with $\varepsilon = 0.02$ (left) and $\varepsilon = 0.01$ (right).

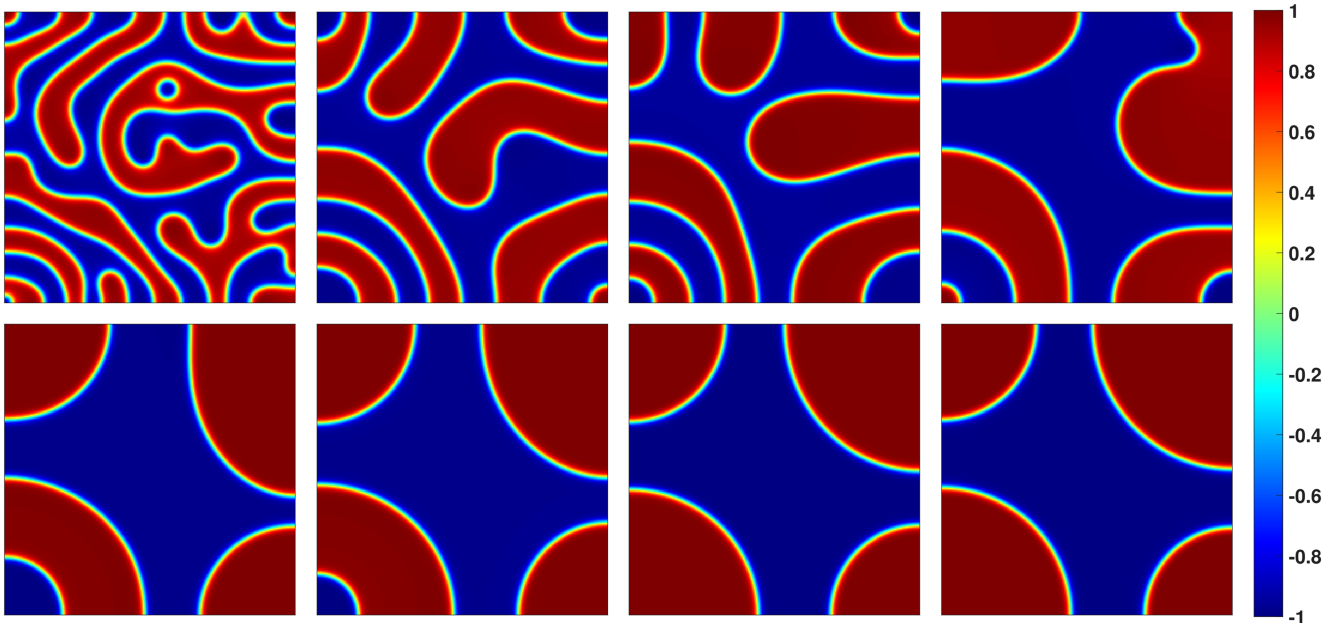


FIGURE 4 | Snapshots of the phase variable during the spinodal decomposition at $t = 0.25, 1, 2, 6, 10, 15, 20$, and 25 (from left to right and top to bottom) by the second-order DRLM scheme (19) with $\bar{\phi}_0 = 0$.

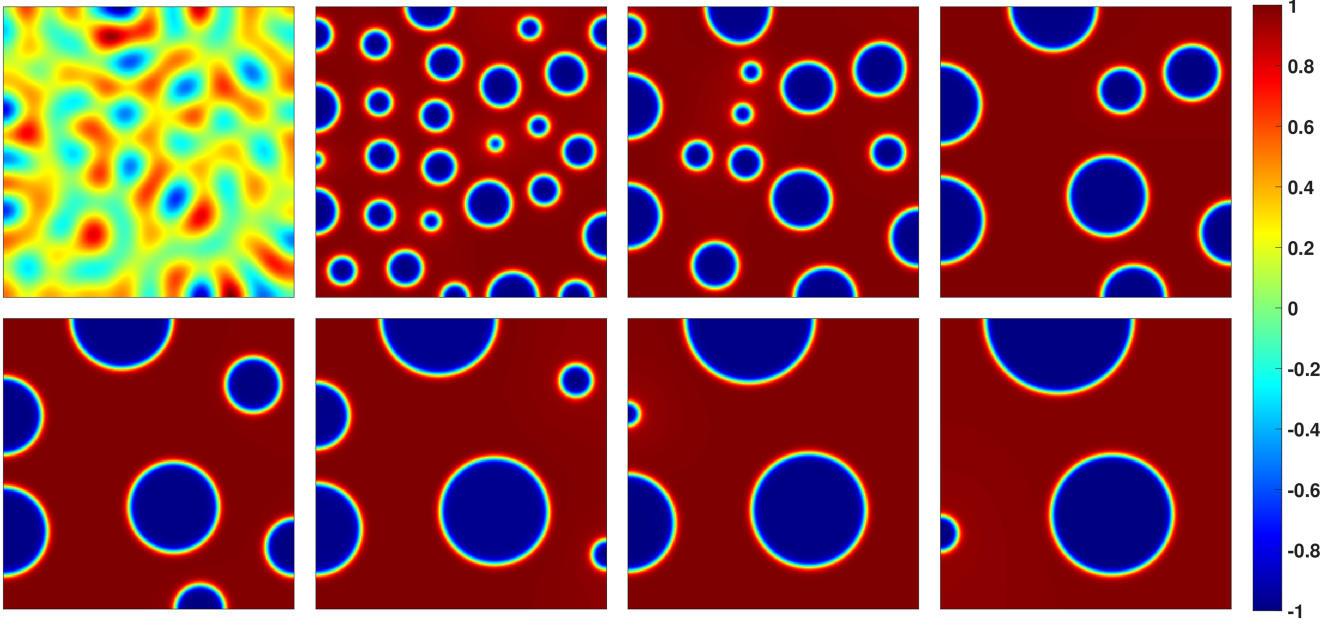


FIGURE 5 | Snapshots of the phase variable during the spinodal decomposition at $t = 0.25, 1, 2, 6, 10, 15, 20$, and 50 (from left to right and top to bottom) by the second-order DRLM scheme (19) with $\bar{\phi}_0 = 0.5$.

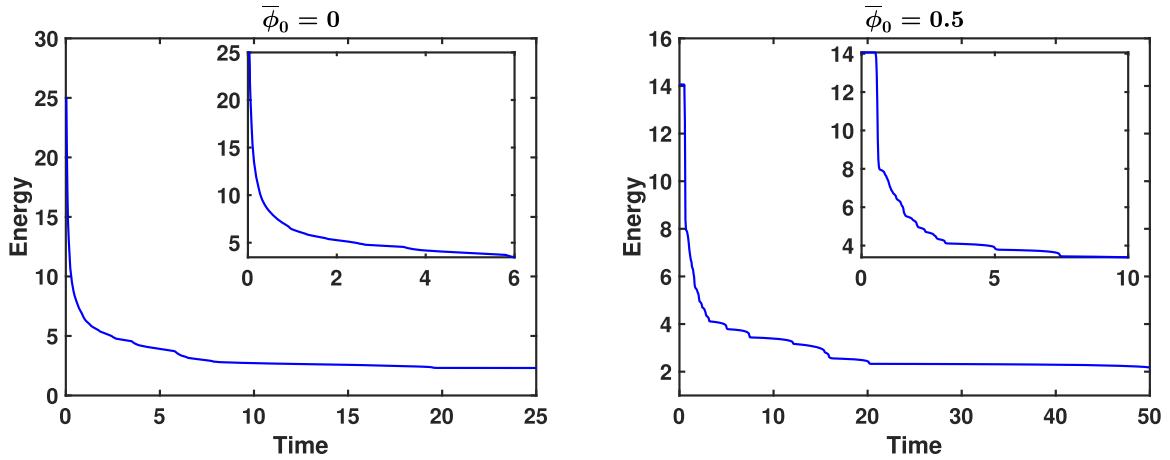


FIGURE 6 | Evolution of the total energy during the spinodal decomposition by the second-order scheme (19) with $\bar{\phi}_0 = 0$ (left) and $\bar{\phi}_0 = 0.5$ (right).

The spatial mesh size and time step size are given by $h = 4 \times 10^{-3}$ and $\tau = 4 \times 10^{-4}$, respectively. Snapshots of the phase evolution with different Reynolds numbers $Re \in \{1, 10, 50\}$, corresponding to $\nu \in \{1, 0.1, 0.02\}$, by the second-order DRLM scheme are presented in Figure 7. In all cases, the droplet first elongates under the effect of gravity, then pinches off and moves downward to the bottom of the domain. As the Reynolds number increases, the pinch-off behavior occurs earlier, and the droplet descends more quickly. Moreover, a circular shape for the droplet is observed at $Re = 1$, while a bean-like shape appears at $Re = 50$, as illustrated in the last subfigure of each row in Figure 7.

We further extend the simulation in three dimensions with the domain $\Omega = (0, 1) \times (0, 1) \times (0, 2)$ and the initial conditions:

$$\phi_0(x, y, z) = \tanh \left(\frac{0.32 - \sqrt{(x - 0.5)^2 + (y - 0.5)^2 + (z - 2.1)^2}}{\epsilon} \right)$$

$$u_0(x, y, z) = 0 \quad (69)$$

We fix $\epsilon = 0.025$ and $Re = 50$ (i.e., $\nu = 0.02$), with other parameters chosen as in the two-dimensional case. Isosurface snapshots

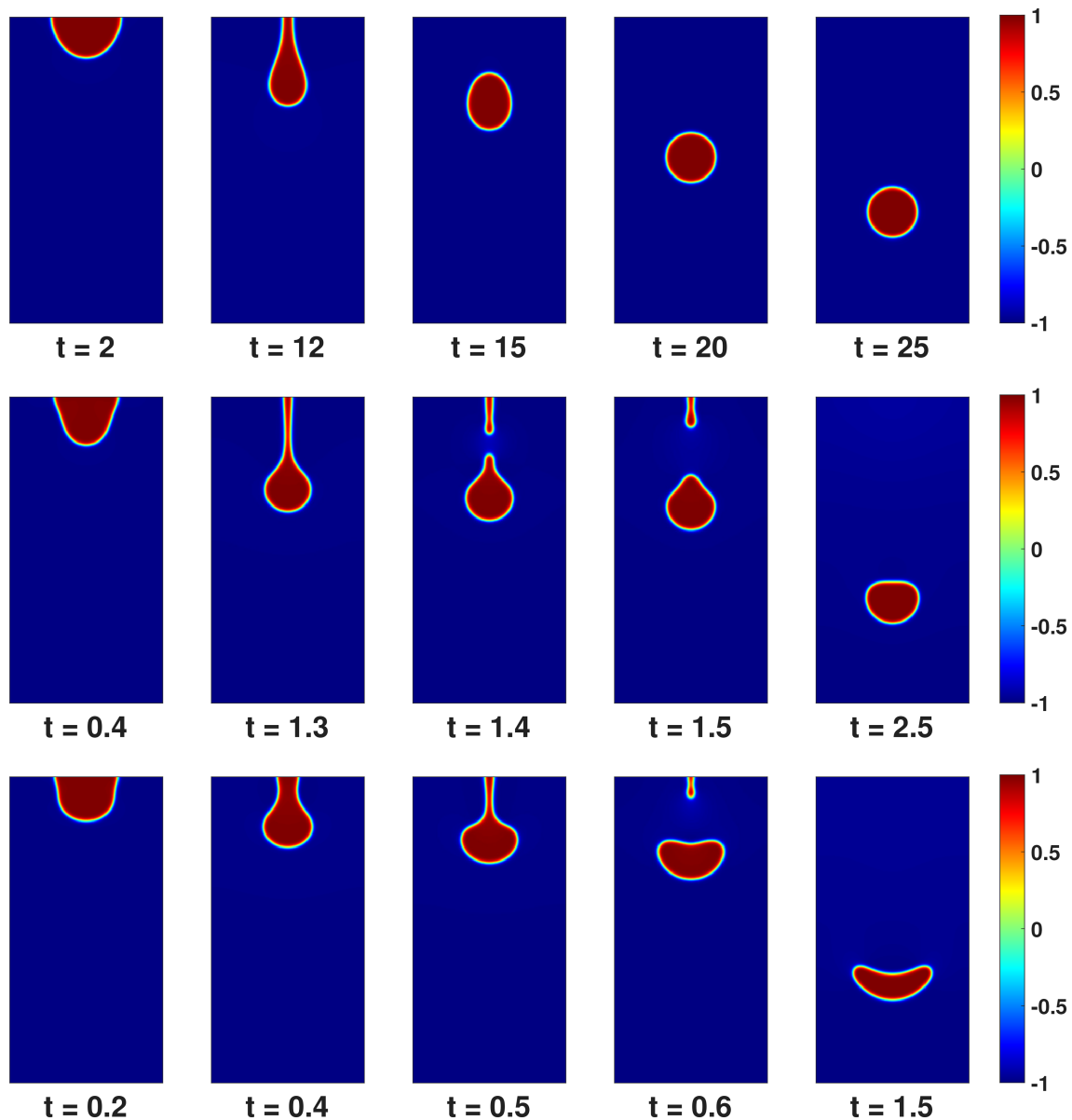


FIGURE 7 | Evolution of the 2D droplet falling at different times with $Re = 1, 10$, and 50 (from top row to bottom row) by the second-order DRLM scheme (19).

of the droplet (i.e., $\{\phi = 0\}$) at different times by the second-order DRLM scheme with $h = 1.25 \times 10^{-2}$ and $\tau = 2 \times 10^{-3}$ are shown in Figure 8. An elongated liquid filament gradually forms and pinches off at around $t = 0.7$, resulting in a bowl-shaped droplet. This behavior is consistent with the two-dimensional simulation, where a similar process produces a droplet with a comparable concave shape.

6 | Conclusions

In this work, we proposed the time-discrete DRLM schemes for solving the CH-NS phase field model of two-phase incompressible fluids with matched density. By introducing two Lagrange multipliers along with their regularization parameters and dynamic equations, the CH-NS equations were transformed into

an equivalent system involving the evolution of the kinetic energy and the nonlinear part of the free energy. First- and second-order DRLM schemes were constructed based on the BDF method with all nonlinear terms treated explicitly, and were shown to satisfy mass conservation and unconditional energy stability. These schemes are efficient and straightforward to implement as they only require the solution of a sequence of linear systems and two nonlinear scalar equations for the Lagrange multipliers at each time step. Various numerical experiments confirmed the theoretical findings and showcased the ability of DRLM schemes in accurately capturing the system's dynamics. Although we observed numerically that large regularization parameters guarantee the uniqueness and accuracy of the Lagrange multipliers, providing theoretical lower bounds for these parameters is still challenging and requires further investigation. Other future directions include fully discrete energy stability and error analysis of the

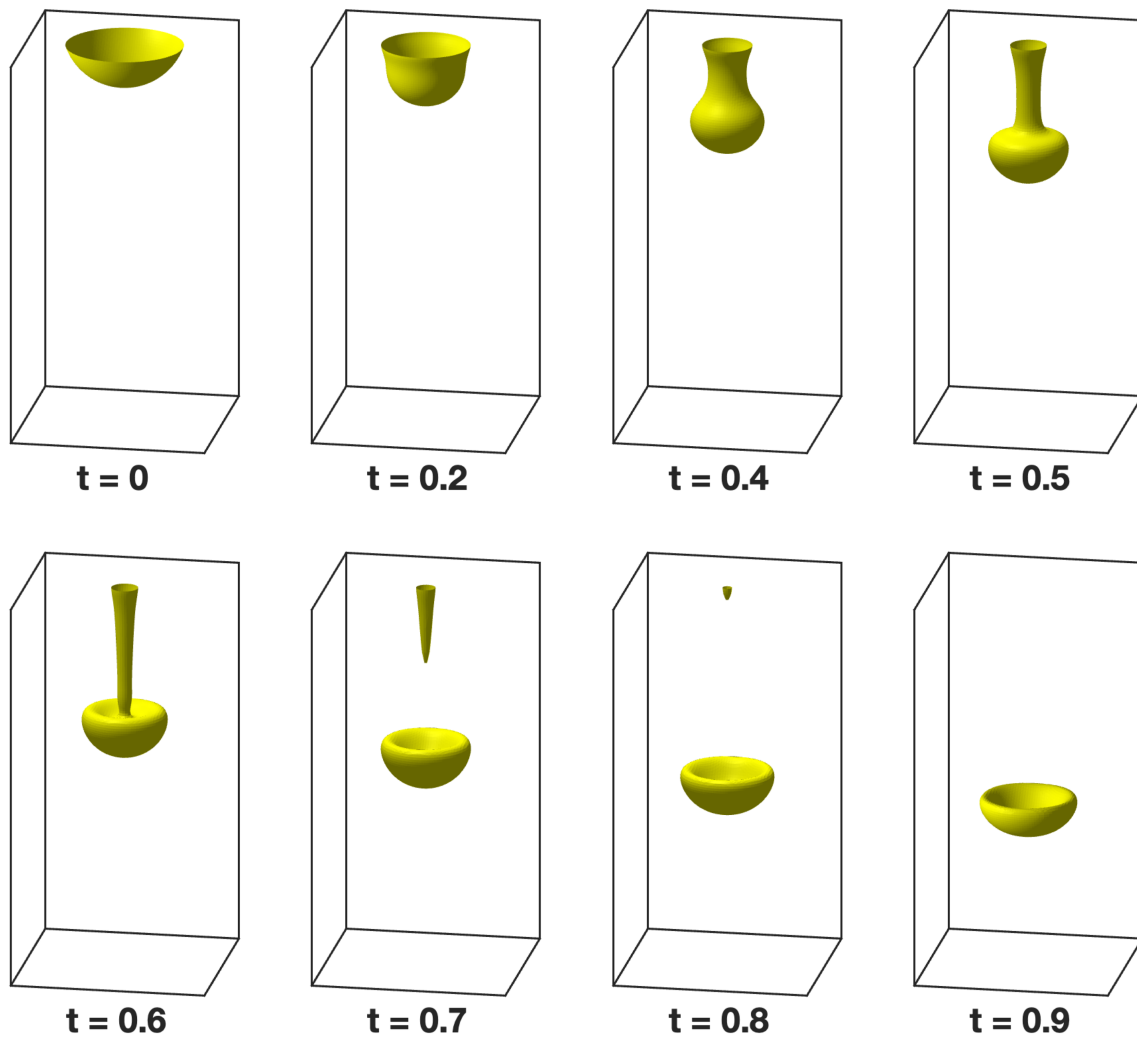


FIGURE 8 | Evolution of the 3D droplet falling at different times with $Re = 50$ by the second-order DRLM scheme (19).

DRLM schemes, as well as extension of the DRLM approach to other types of PDEs, such as magnetohydrodynamic equations. We refer to [19, 55] for the convergence analysis of numerical schemes based on the finite element or discontinuous Galerkin method for the CH-NS equations.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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